Cluster analysis

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Chapters for cluster analysis

ISLR: Chapter 12: Unsupervised Learning

- Section 12.4: Clustering Methods
  - Section 12.4.1: K-Means Clustering
  - Section 12.4.2: Hierarchical Clustering

R Book: Chapter 25: Multivariate Statistics

- Section 25.3: Cluster Analysis
  - Section 25.3.1: Partitioning
  - Section 25.3.2: Taxonomic Use of K-means
- Section 25.4: Hierarchical Cluster Analysis

Topics for this lecture

- Cluster analysis
- Similarity and distance
- K-means clustering
- Partitioning Around Medoids (PAM) clustering
- Hierarchical clustering

Cluster analysis

- Objects belong to some groups or classes
- Cluster analysis is so-called *Unsupervised learning* of groups
- Neither the identity nor the number of groups is known
- Can be a non-parametric (algorithmic) data-driven search - today
- Or statistical grounded through semi-parametric mixture models - advanced
- Later in the course, we will return to *supervised learning* where the identities and number of groups are known (discriminant analysis and classification)

Groups in data

- A data set with n cases (rows) and p variables (columns)

$$\mathbf{X} = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix}$$

- We look for groups in the data cases
  - Cases in the same group are «similar»
  - Cases in different groups are «different»
- Grouping or similarity is relative to some **similarity** or **distance metric**
  - What do we mean by the «distance» between two points?

Motivation

- **Explorative**
  - A way to «inspect» large data sets
  - May give you some ideas/hypotheses
  - May reveal outliers or severe errors in data
- **Down-sampling**
  - Each cluster center is a representative of its «region» in the data

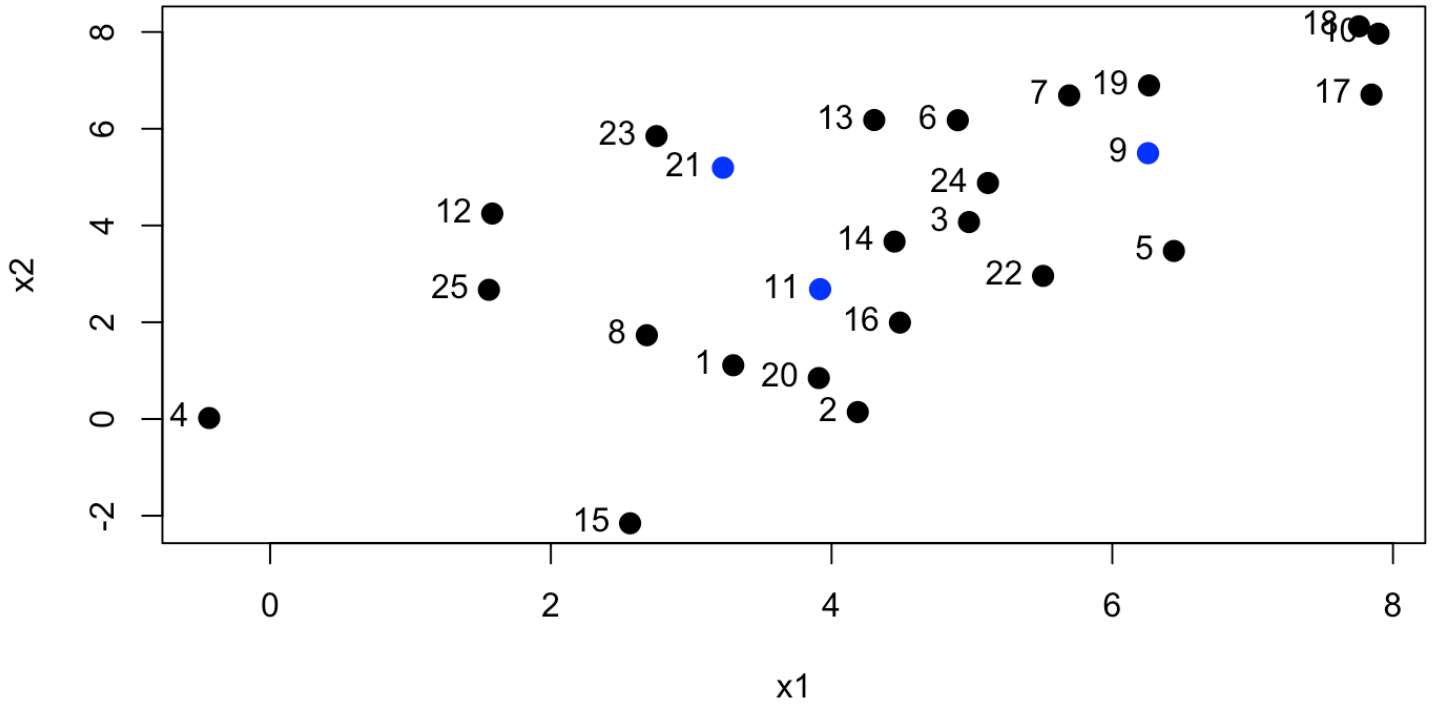
Measure of similarity/dissimilarity

We need measures of “distance” between observations

- Euclidean distance
- Manhattan distance
- Minkowski distance
- Hamming distance (for binary variables)
- Others

See e.g `?dist` for R help

A simulated example - Distance between observations?



Euclidean distance

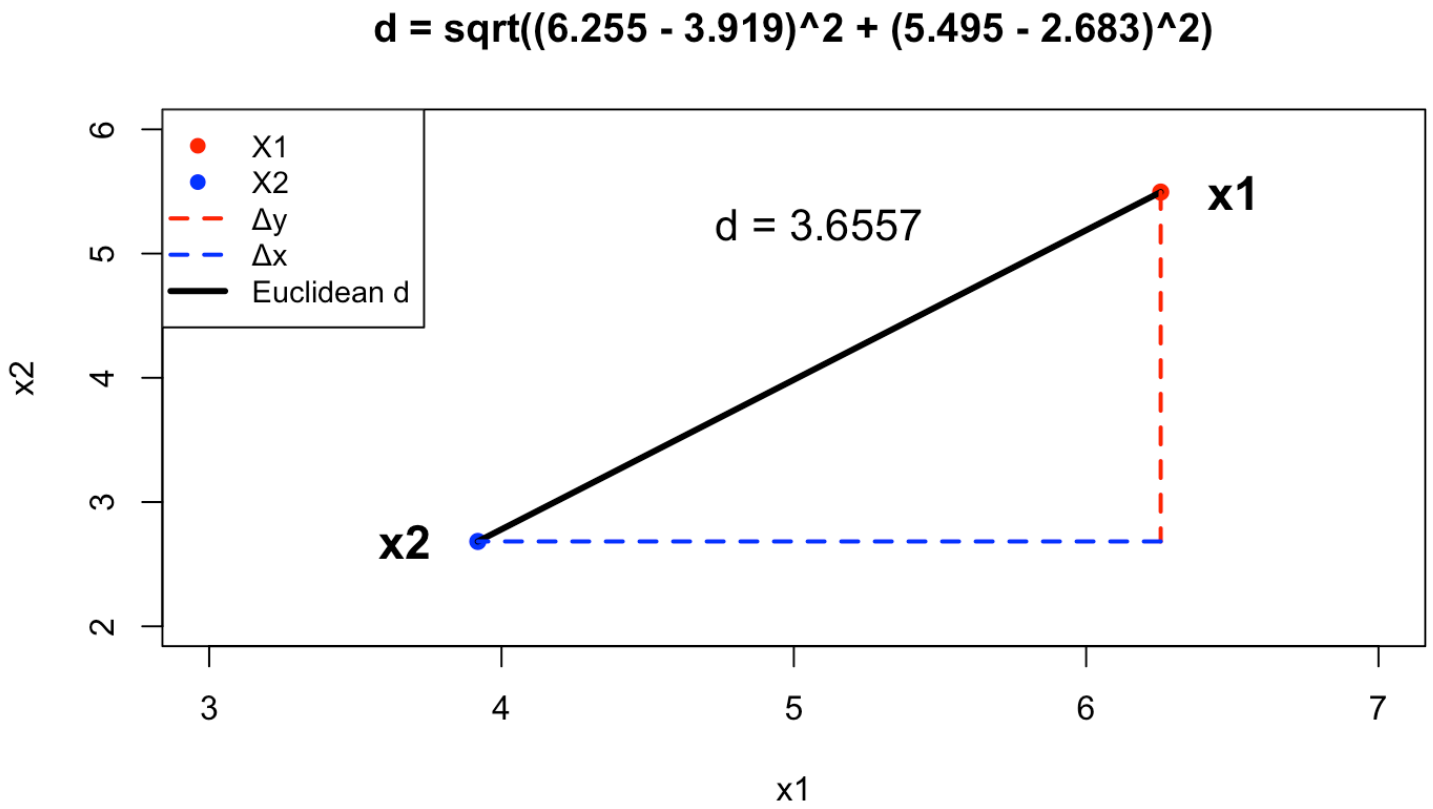
Euclidean distance is shortest distance on a map

E.g. distance between two observations (x_{11}, x_{21}) and (x_{12}, x_{22})

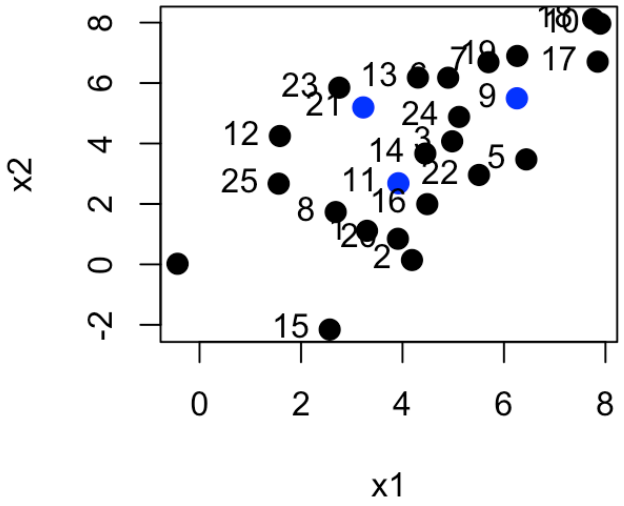
$$D_e = \sqrt{(x_{11} - x_{12})^2 + (x_{21} - x_{22})^2}$$

Euclidean distance: example

Assume $x_1 = (6.255, 5.495), x_2 = (3.919, 2.683)$



For blue observations no (9, 11 and 21)

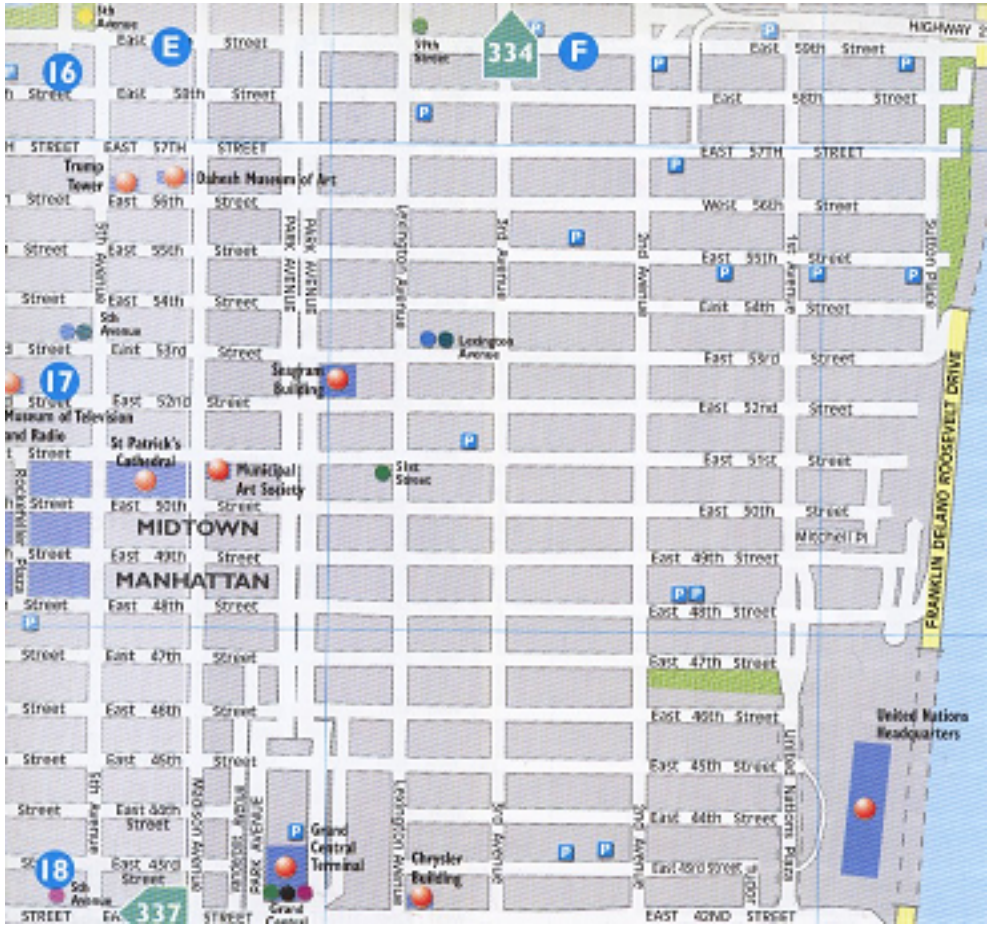



```
dist(mypoints, method ="euclidean")
```

```
##           9       11
## 11 3.655337
## 21 3.043111 2.604636
```

Manhattan distance

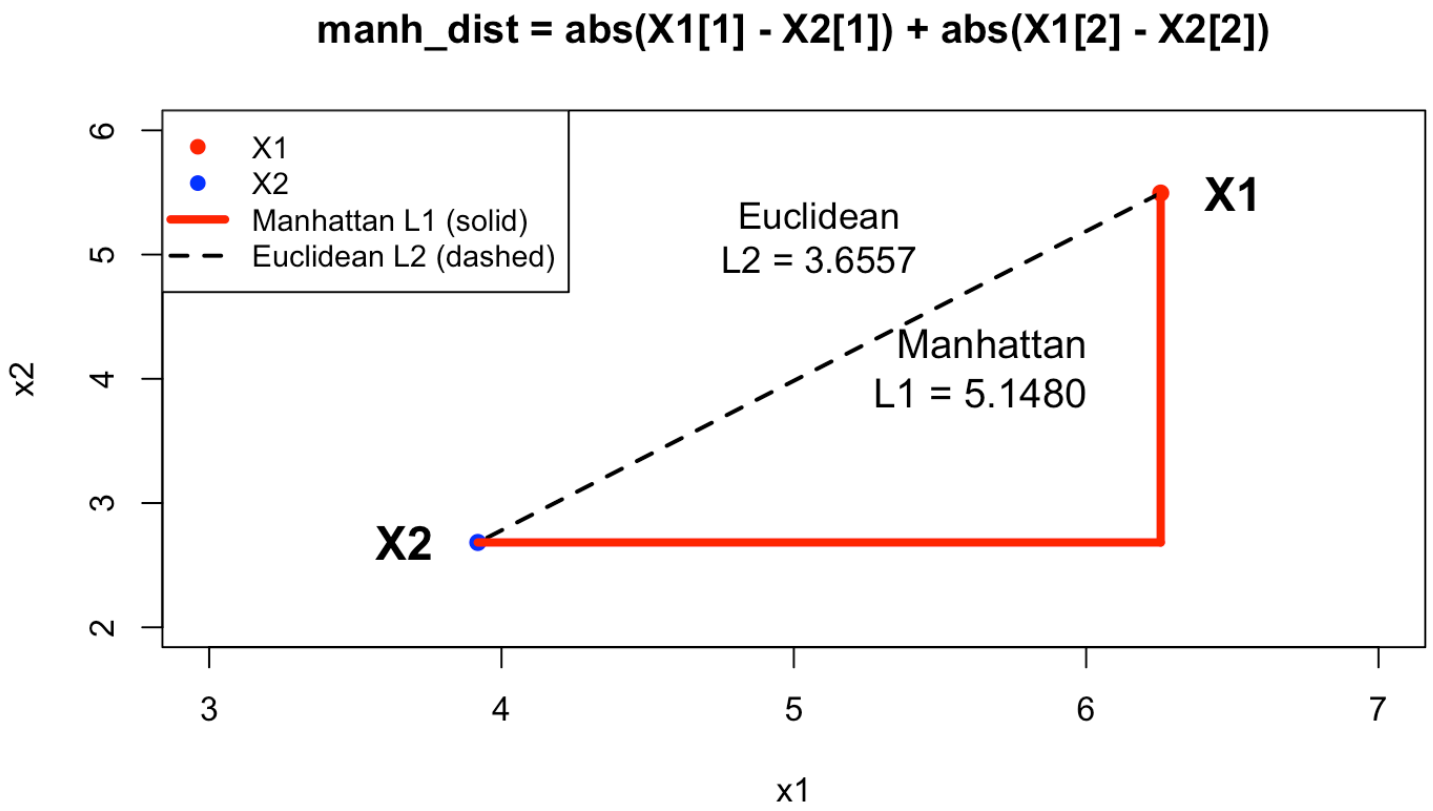
“Manhattan distance”” is like the walking distance between two shops on Manhattan



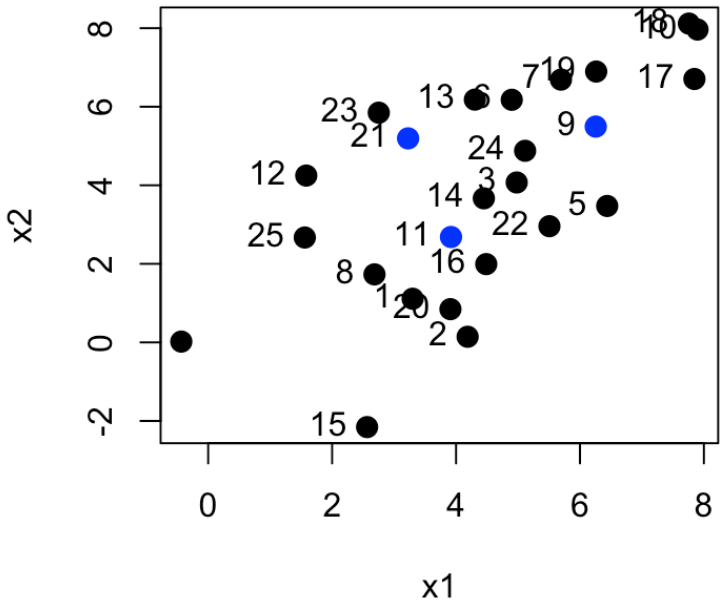
Manhattan distance, continued

Manhattan distance between two observations
$x_1 = (6.255, 5.495), x_2 = (3.919, 2.683)$

$$D_m = |x_{11} - x_{12}| + |x_{21} - x_{22}|$$



Manhattan distance for blue points:




```
##           9       11
## 11 5.147513
## 21 3.328770 3.203190
```

Minkowski distance

For two points $\mathbf{x}_1 = (x_{11}, \ldots, x_{1p})$ and $\mathbf{x}_2 = (x_{21}, \ldots, x_{2p})$, the **Minkowski distance** of order $r \geq 1$ is

$$d_r(\mathbf{x}_1, \mathbf{x}_2) = \left(\sum_{j=1}^{p} |x_{1j} - x_{2j}|^r \right)^{1/r}.$$

- **Manhattan distance** ($r = 1$): $d_1(\mathbf{x}_1, \mathbf{x}_2) = \sum_{j=1}^{p} |x_{1j} - x_{2j}|$
- **Euclidean distance** ($r = 2$): $d_2(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{\sum_{j=1}^{p} (x_{1j} - x_{2j})^2}$
- Larger r values give more weight to large coordinate differences.

Hamming distance

For two **binary** vectors $\mathbf{x}_1, \mathbf{x}_2 \in \{0,1\}^p$ of the same length, the **Hamming distance** counts how many positions differ:

$$d_H(\mathbf{x}_1, \mathbf{x}_2) = \#\{j : x_{1j} \neq x_{2j}\}.$$

Example (differences highlighted in red):

$$\mathbf{x}_1 = (1, 0, 1, 0), \quad \mathbf{x}_2 = (0, 0, 0, 0)$$

Positions 1 and 3 differ, so $d_H(\mathbf{x}_1, \mathbf{x}_2) = 2$.

- Used for **categorical** or **binary** variables (e.g., gene sequences, binary features).
- Related to the number of bit flips needed to transform one vector into the other.

Iris data

Goal: Group together objects that are similar

Iris versicolor

Iris virginica

The famous "Iris" data set

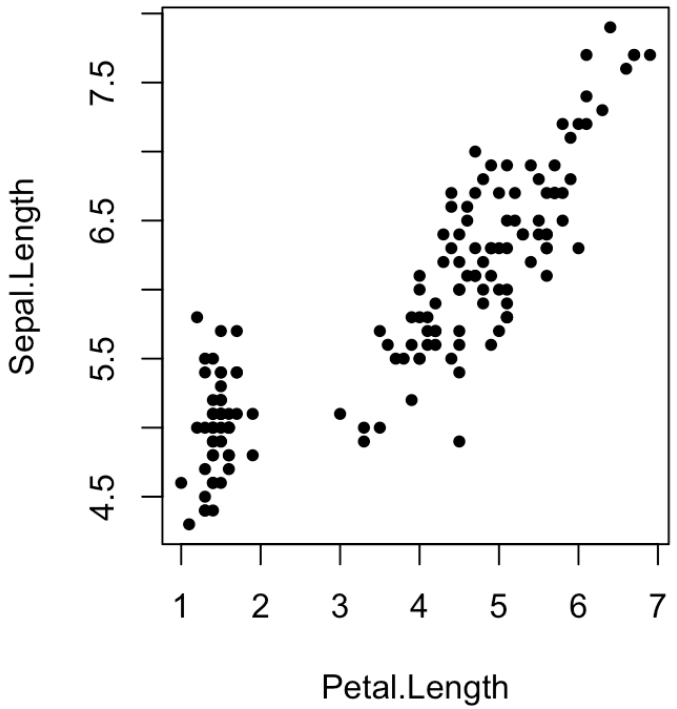
- This famous (Fisher's or Anderson's) iris data set gives the measurements in centimeters of the variables X1=sepal length, X2=sepal width, X3=petal length and X4=petal width, respectively.

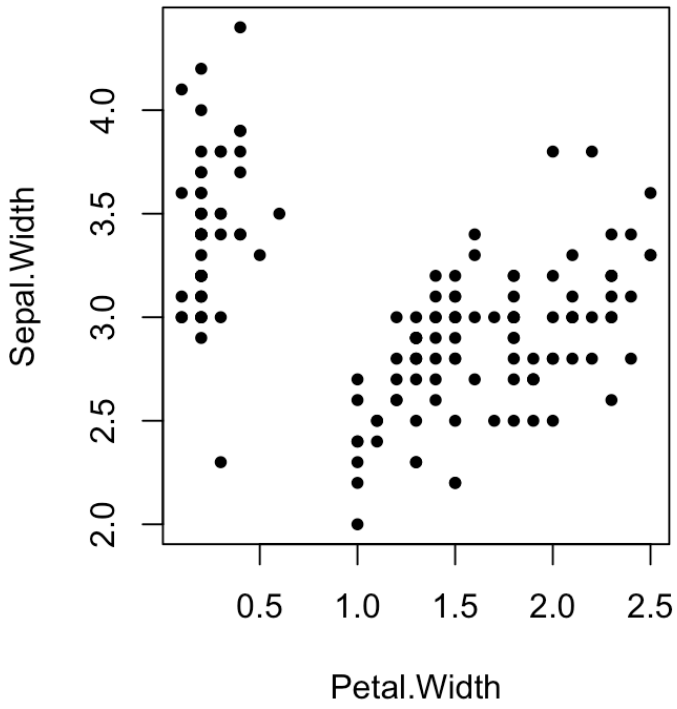
```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
## 1            5.1         3.5          1.4         0.2     setosa
## 2            4.9         3.0          1.4         0.2     setosa
## 3            4.7         3.2          1.3         0.2     setosa
## 55           6.5         2.8          4.6         1.5 versicolor
## 56           5.7         2.8          4.5         1.3 versicolor
## 57           6.3         3.3          4.7         1.6 versicolor
## 125          6.7         3.3          5.7         2.1  virginica
## 126          7.2         3.2          6.0         1.8  virginica
```

- Question: Are there many species of iris? which are which? (if we did not know the truth)

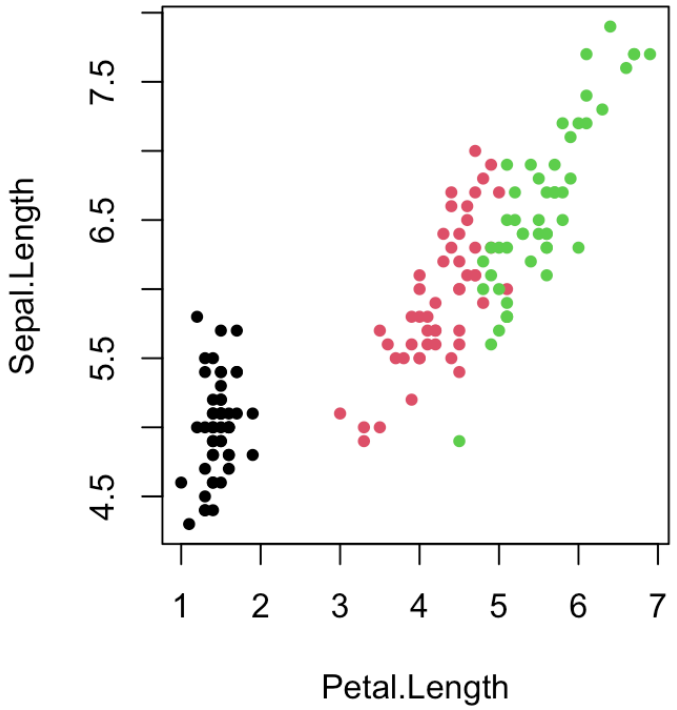
Sepal vs petal length/width

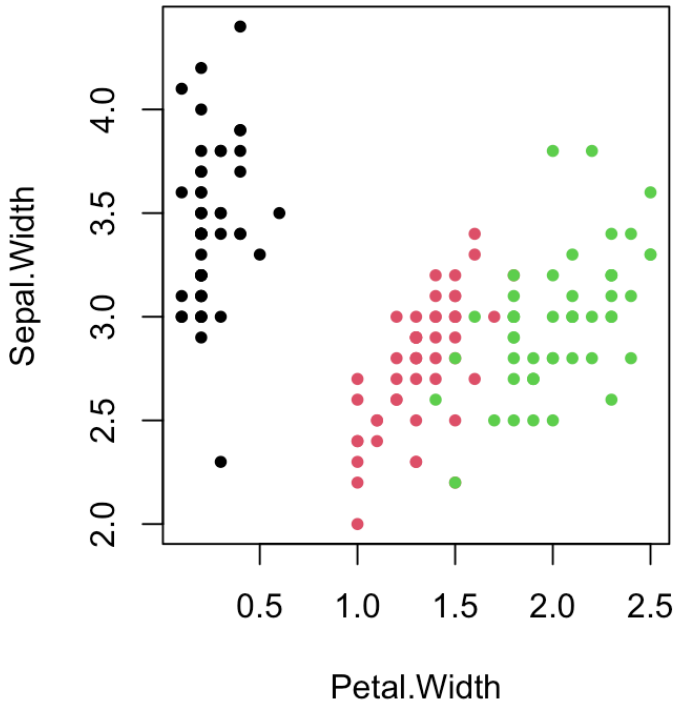
- Are there detectable clusters in the data?





For these data we know…





K-means clustering

- Distribute the observations into K clusters so that the **within-cluster** dissimilarity (from **Euclidean distance**) is minimized.
- For cluster C_k, the within-cluster sum of squared distances is

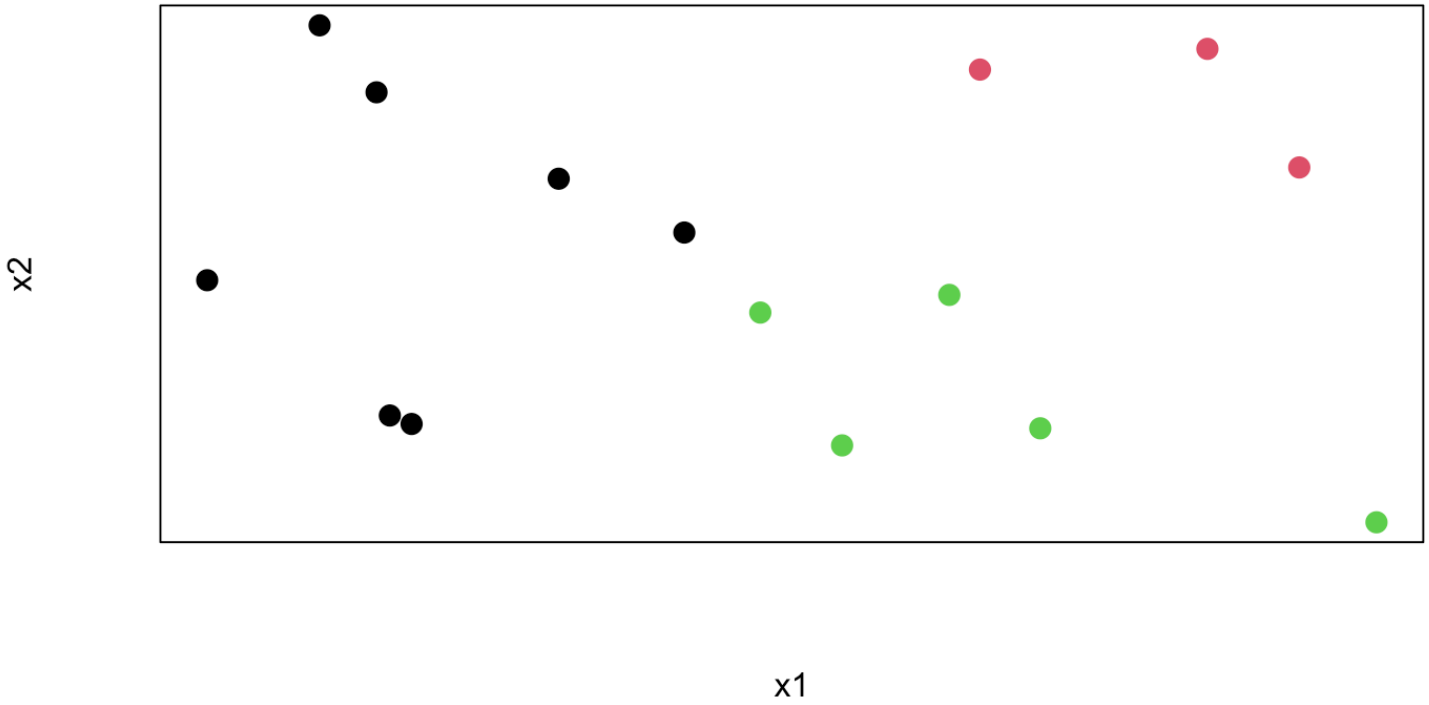
$$W(C_k) = \sum_{\mathbf{x}_i \in C_k} d^2(\mathbf{x}_i, \boldsymbol{\mu}_k),$$

  where $d(\cdot, \cdot)$ is the distance (here Euclidean) and $\boldsymbol{\mu}_k$ is the centroid of C_k.
- The k-means objective is to find clusters $C_1, \ldots, C_K$ such that

$$\text{WCSS}_K = \sum_{k=1}^{K} W(C_k) \to \min.$$

K-means illustrated, I

- Two variables, x1 and x2, and $K = 3$ clusters



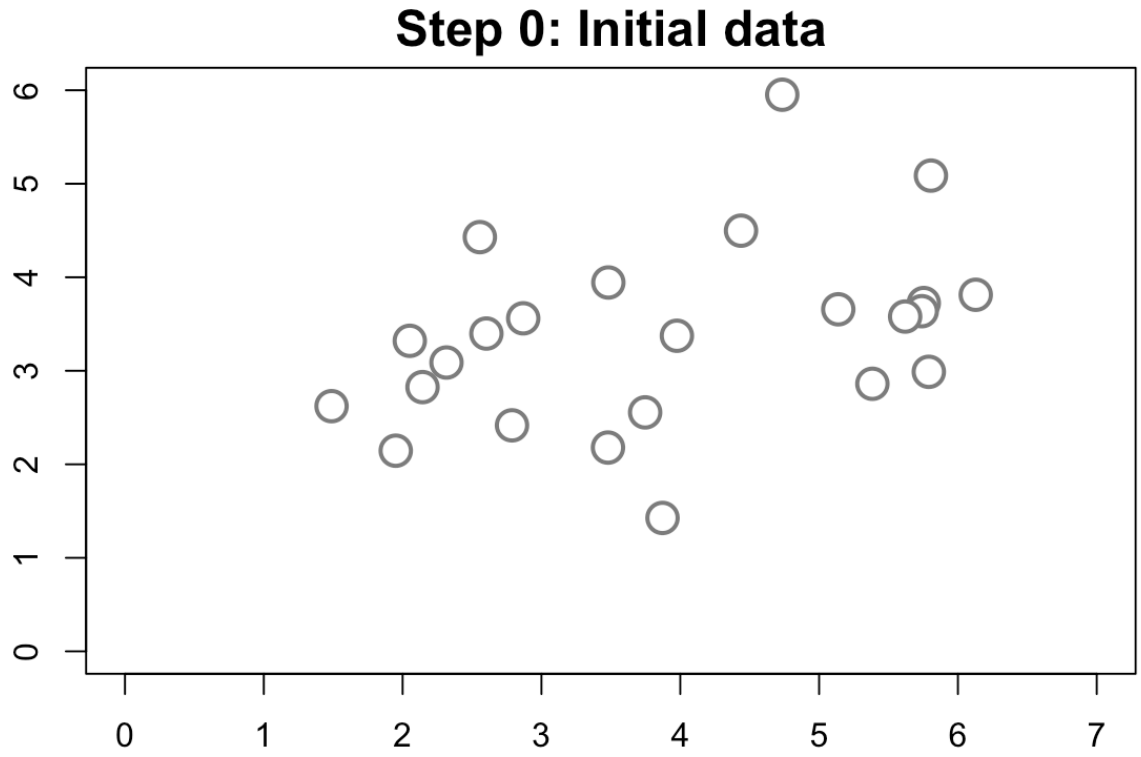
K-means clustering algorithm

1. Choose the number of clusters K and select K initial centroids (typically random observations).
2. **Assignment step**: Assign each observation to the cluster with the closest centroid (Euclidean distance).
3. **Update step**: Recompute each centroid as the mean of the observations currently in that cluster.
4. Repeat steps 2–3 until the cluster assignments no longer change.

- Sensitive to the random starting centroids → may converge to a **local** minimum; therefore run k-means several times and keep the best solution.

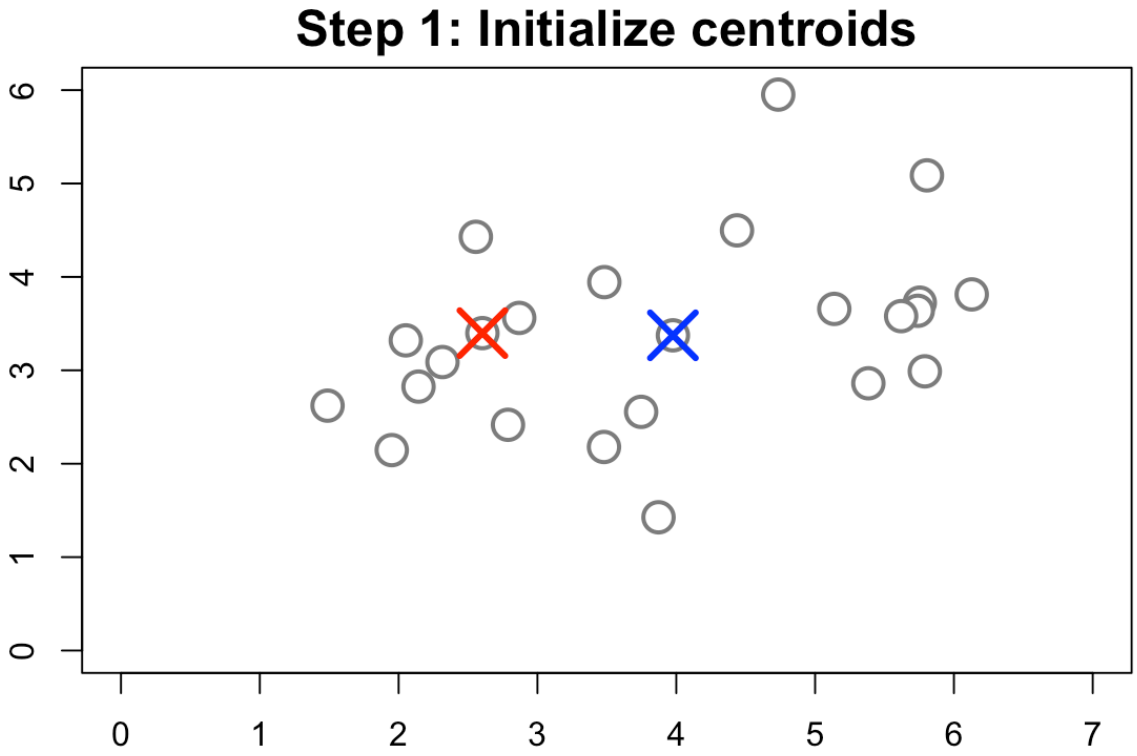
K-means: Step 0

Initial data, no clusters yet.



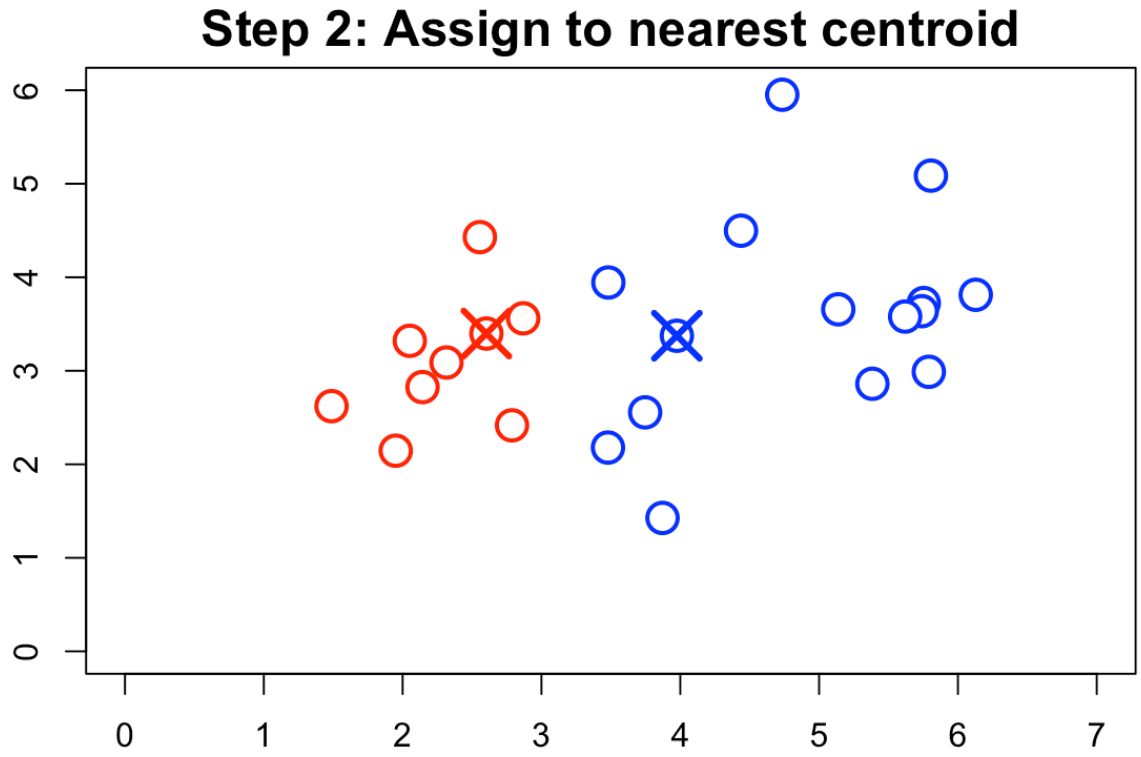
K-means: Step 1

Initialize: Randomly place $K = 2$ centroids.



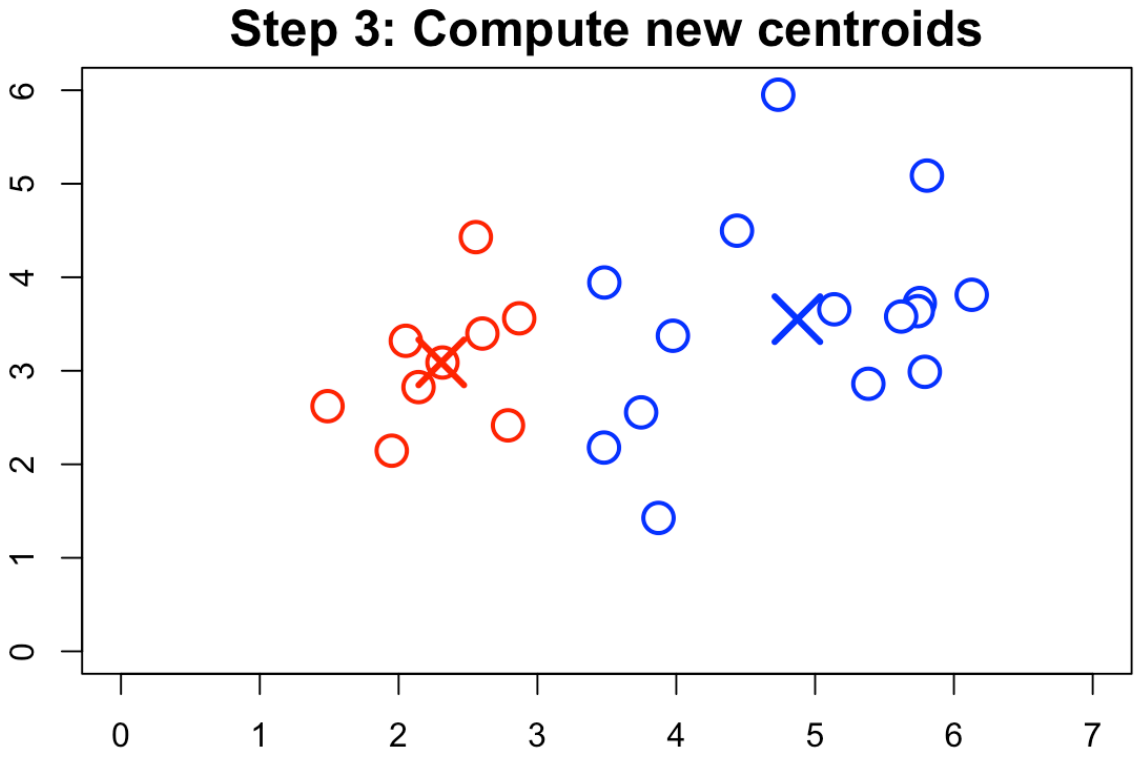
K-means: Step 2

Assign each observation to the nearest centroid.



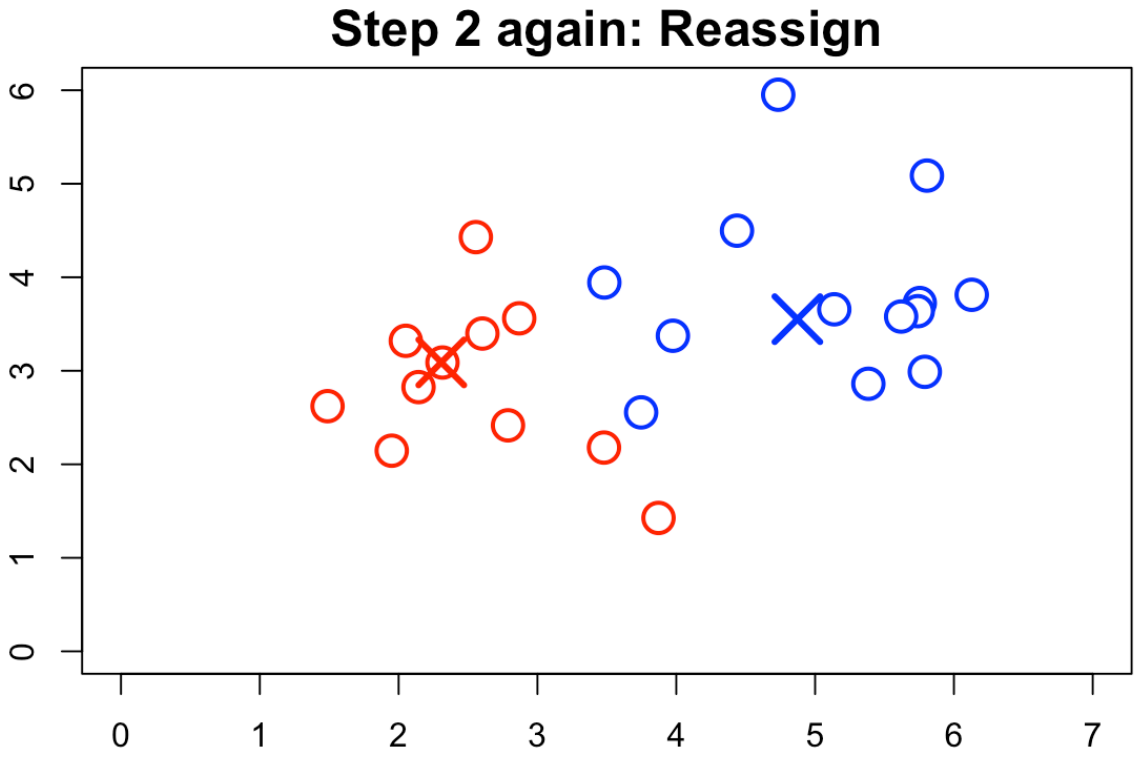
K-means: Step 3

Update: Compute new centroids as the mean of assigned points.



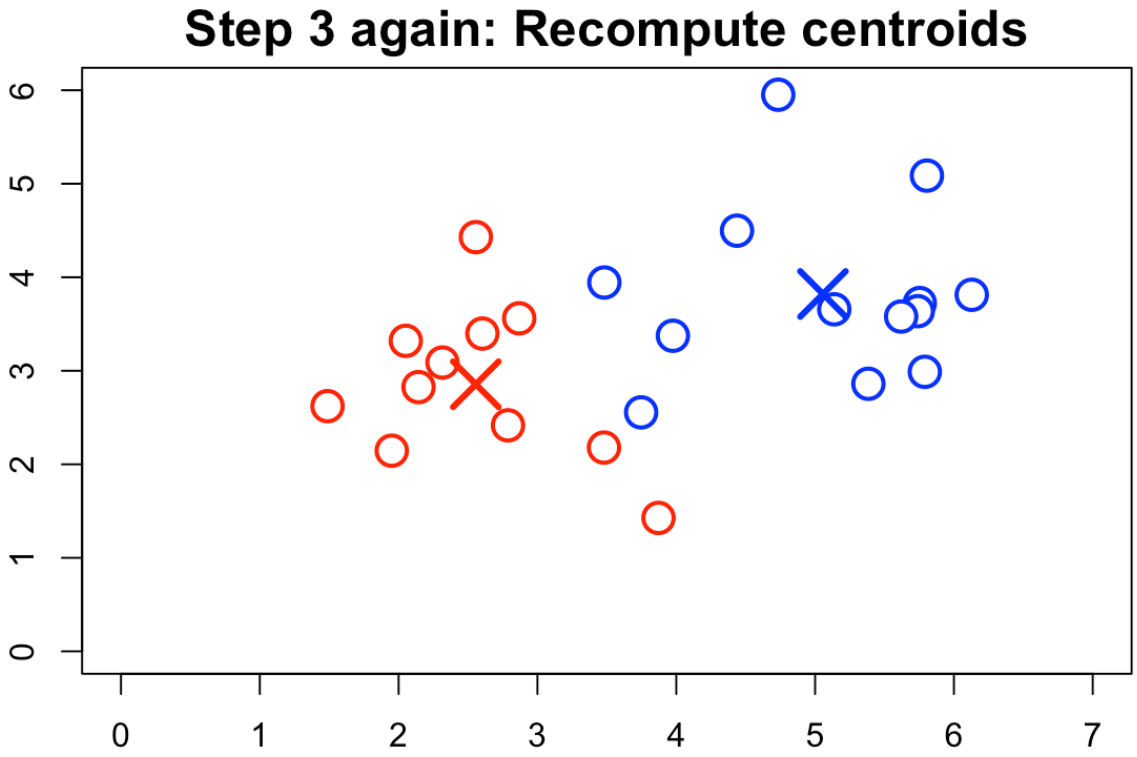
K-means: Step 2 again

Repeat Step 2 with updated centroids. If nothing changes we are done



K-means: Step 3 again

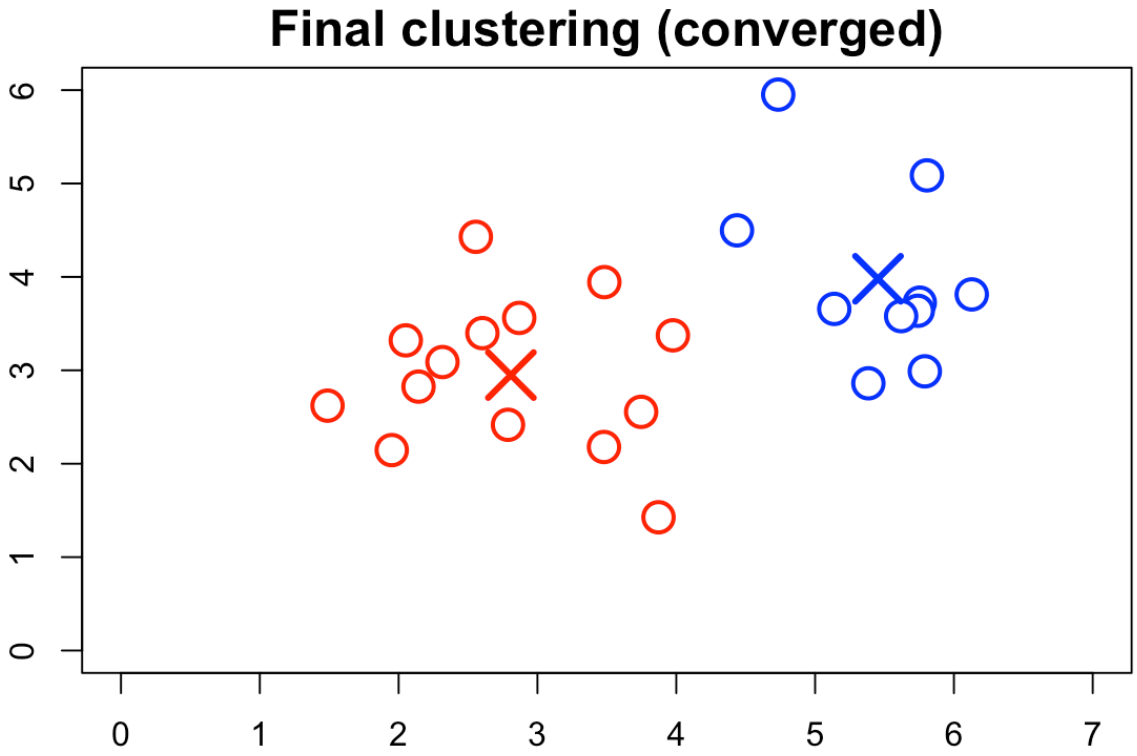
Repeat Step 3: recompute centroids.



K-means: Convergence

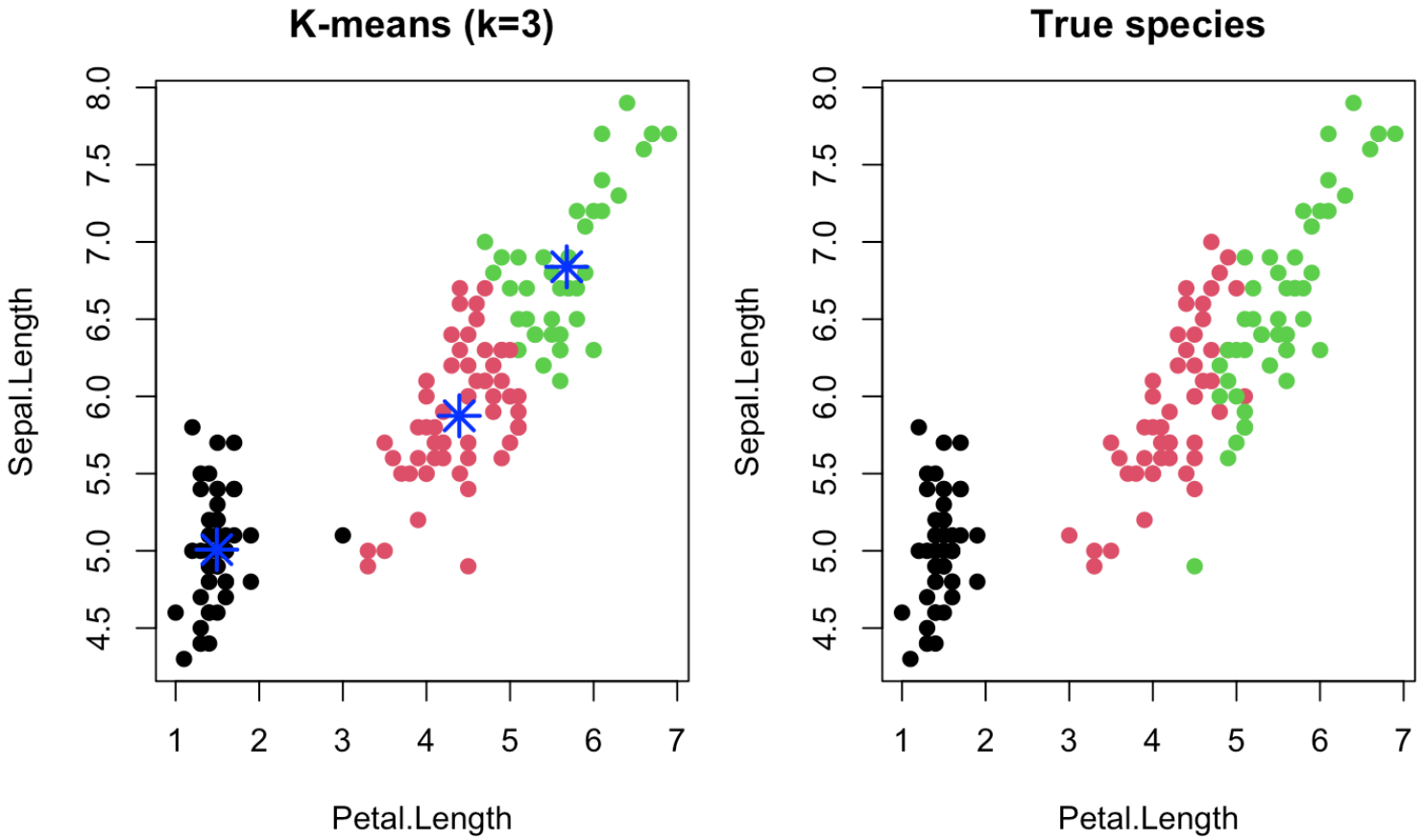
If assignments not changing, the algorithm **converged**. In R:

```
km_final <- kmeans(X, centers = 2, nstart = 100)
```



K-means on the iris data (R-code)

```
df <- data.frame(x=iris[,3], y=iris[,1])
kmeans_result <- kmeans(df, centers = 3,nstart = 100)
```



K-means sums of squares

```
kmeans_result$size # Cluster Sizes
```

```
## [1] 51 41 58
```

```
kmeans_result$withinss # Within Cluster Sum of Squares
```

```
## [1]  9.893725 20.407805 23.508448
```

```
kmeans_result$tot.withinss # Total Within Sum of Squares
```

```
## [1] 53.80998
```

Interpretation: - `size`: number of observations in each cluster - `withinss`: sum of squared distances within each cluster - `tot.withinss`: total variation within clusters (lower is better)

K-means on the full iris data (R-code)

```
df <- iris[,1:4]
kmeans_result <- kmeans(df, centers = 3,nstart = 100)
```

K-means sums of squares

```
kmeans_result$size          # Cluster Sizes
```

```
## [1] 62 38 50
```

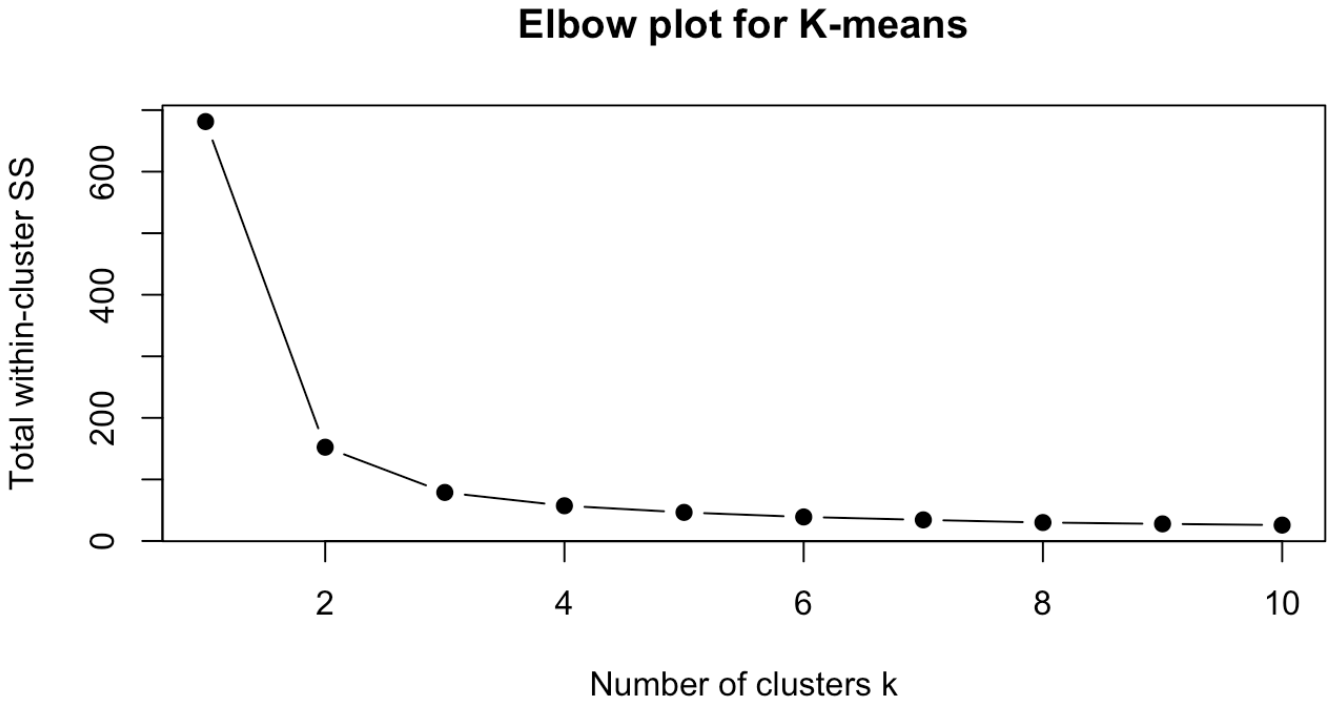
```
kmeans_result$withinss      # Within Cluster Sum of Squares
```

```
## [1] 39.82097 23.87947 15.15100
```

```
kmeans_result$tot.withinss  # Total Within Sum of Squares
```

```
## [1] 78.85144
```

Choosing K: Elbow Plot



Look for the "elbow" where adding more clusters gives diminishing returns.

AIC and BIC for K-means

WCSS_K = `km$tot.withinss` be the residual sum of squares and $\hat{\sigma}^2_K = \text{WCSS}_K/n$ the variance.

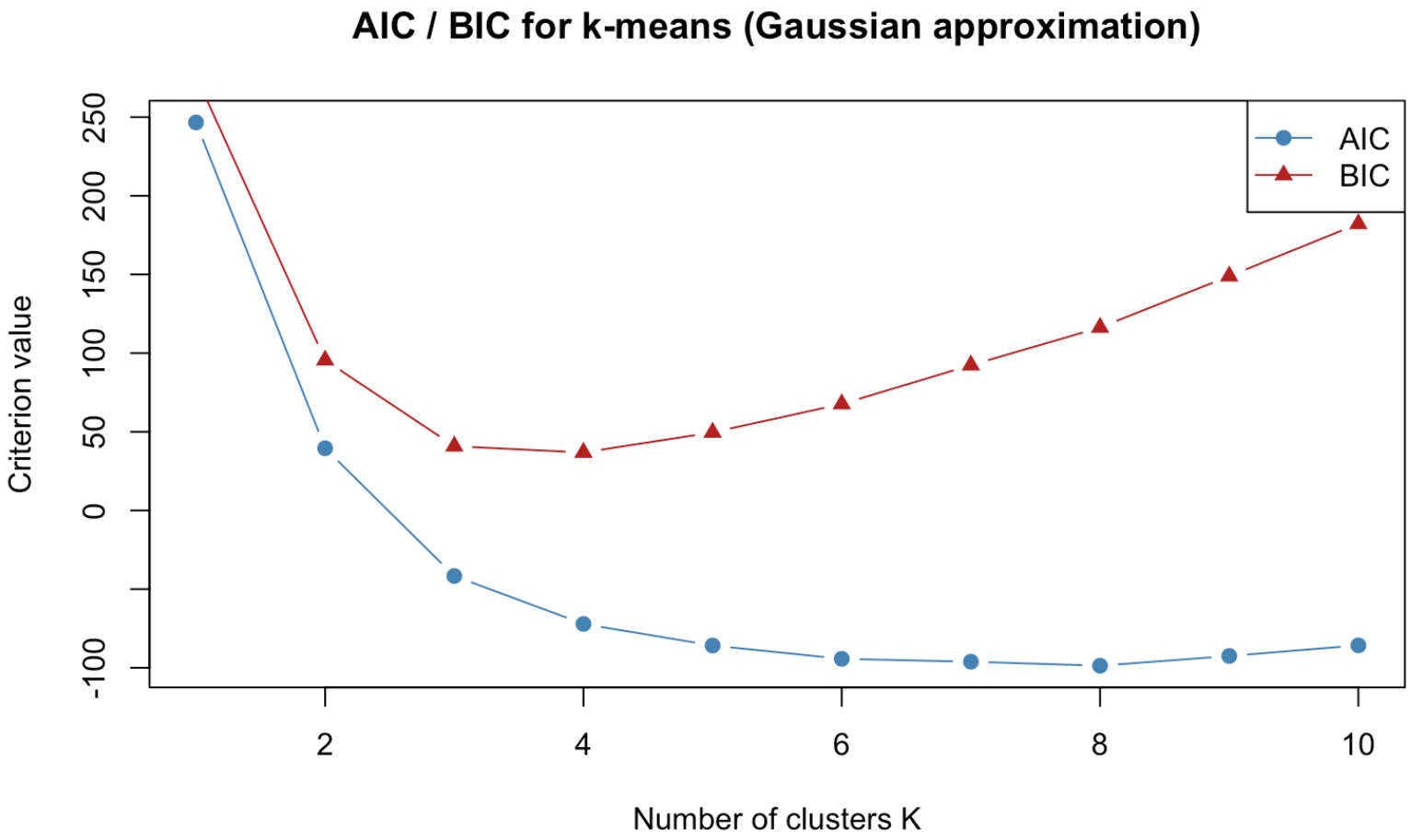
Number of parameters: $\text{params}_K = K \times p + 1$ (K of p-dimensional cluster means plus one common variance). Then

$$\text{AIC}(K) = n \log\left(\frac{\text{WCSS}_K}{n}\right) + 2\,\text{params}_K\,,$$

$$\text{BIC}(K) = n \log\left(\frac{\text{WCSS}_K}{n}\right) + \log(n)\,\text{params}_K\,.$$

- The term $n \log(\text{WCSS}_K/n)$ plays the role of a (negative) log-likelihood based on the residual sum of squares.
- The penalty terms $2\,\text{params}_K$ (AIC) and $\log(n)\,\text{params}_K$ (BIC) discourage using too many clusters.

Choosing K: AIC and BIC



Partitioning Around Medoids (PAM)

- The `kmeans()` function in always uses **Euclidean** distances.
- The `pam()` function (package `cluster`) is similar to k-means, but
  - Uses **medoids** instead of centroids (an actual data point closest to the center).
  - You can supply any distance **matrix** here.

$$W_k = \sum_{i \in C_k} d(\mathbf{x}_i, \mathbf{m}_k) \quad \text{for cluster } C_k$$

Find clusters $C_1, \ldots, C_K$ such that $\sum_{k=1}^{K} W_k \rightarrow \min$

PAM with custom distance matrix

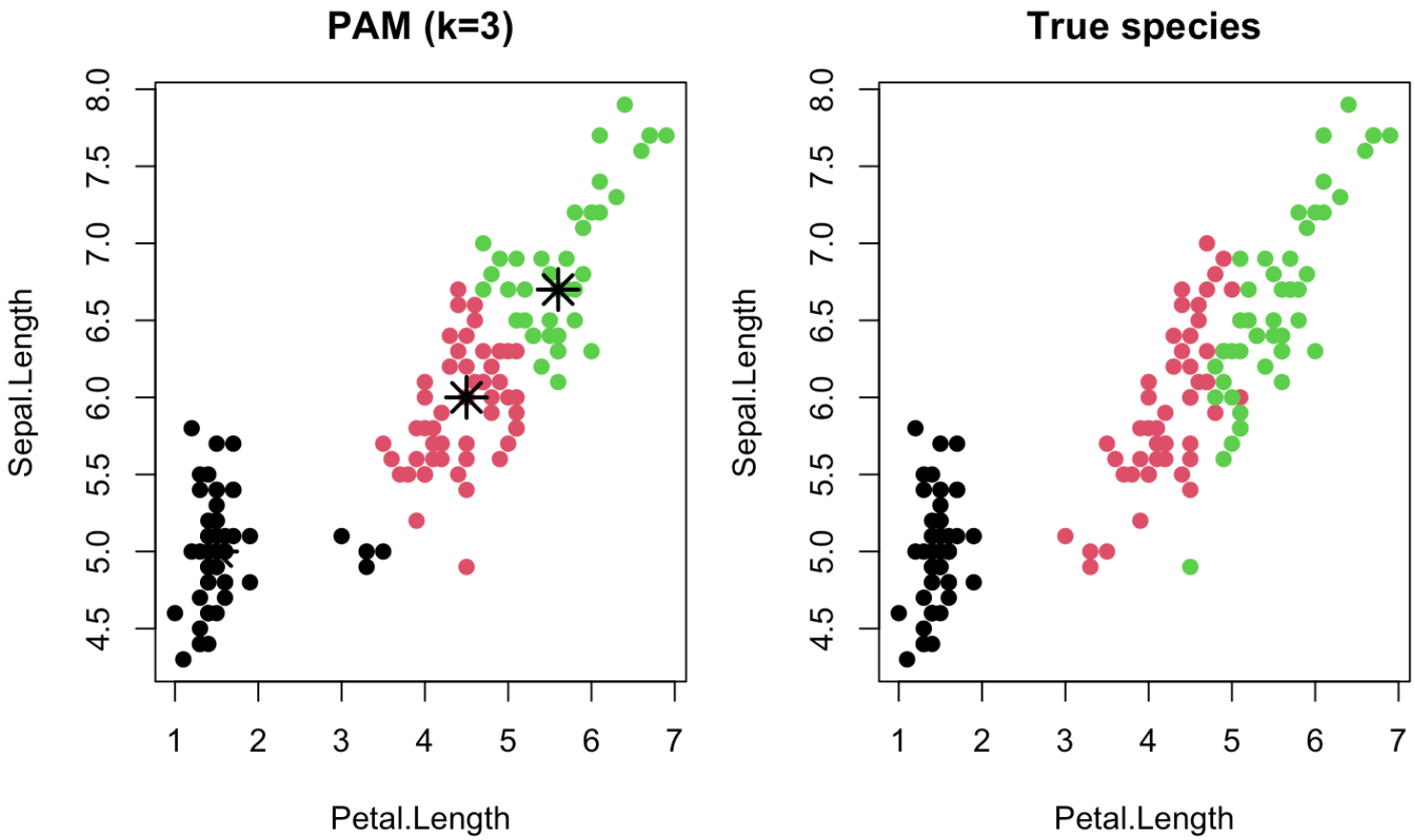
PAM can work with **any dissimilarity matrix**, not just Euclidean distance.

```
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.4.3
```

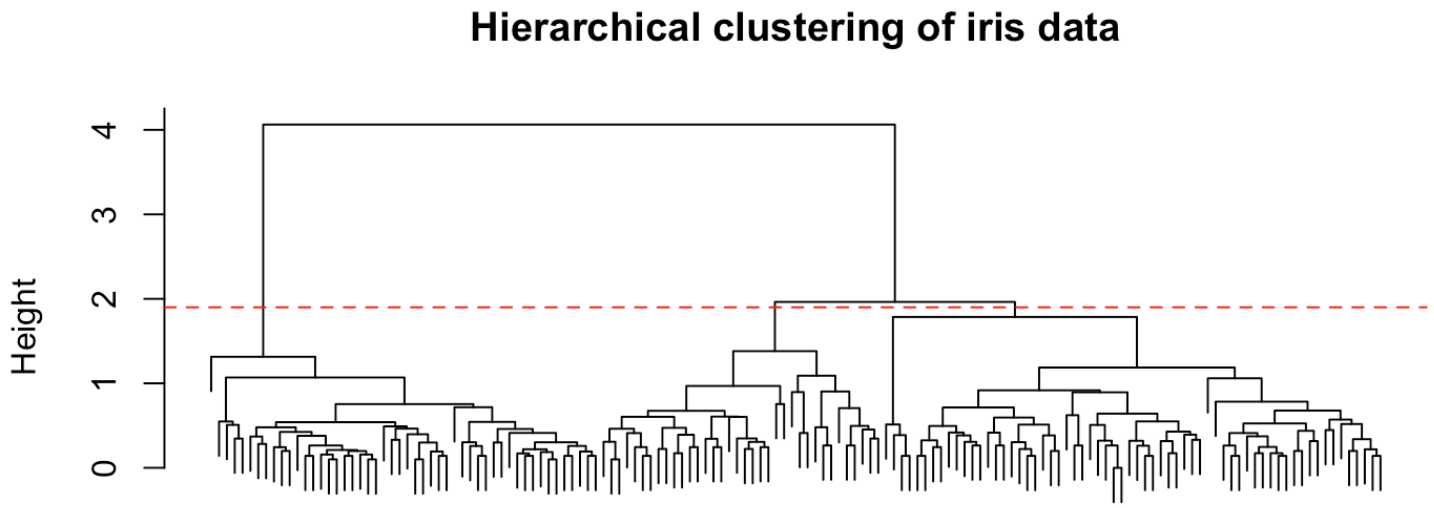
```
set.seed(123)

# Use iris data
df <- iris[, c(3, 1)]# Petal.Length, Sepal.Length
pam_manhattan <- pam(df, k = 3, metric = "manhattan",nstart = 100)

# Compute Minkowski distance matrix
dist_minkowski <- dist(df, method = "minkowski")
# Run PAM on the distance matrix
pam_minkowski <- pam(dist_minkowski, k = 3, diss = TRUE,nstart = 100)
```

PAM clustering on iris data



Hierarchical clustering

- Builds a hierarchical distance structure – dendrogram (tree)
  - We often plot this dendrogram
- Clustering into distinct groups by cutting the dendrogram
- Requires we define distance between clusters (linkage function)



Hierarchical clustering algorithm

Each of the n observations is a point in a p-dimensional space

Agglomerative (bottom-up) algorithm to build a dendrogram:

1. **Initialize**: Start with n clusters, with single observations.
2. **Compute distances**: Calculate the distance between every pair of clusters using a chosen **distance metric**.
3. **Merge**: Find the two clusters with the smallest distance and merge them into a single cluster.
4. **Update distances**: Recompute distances between the new cluster and all remaining clusters using a **linkage function**
5. **Repeat**: Go back to step 3 until all observations belong to a single cluster.

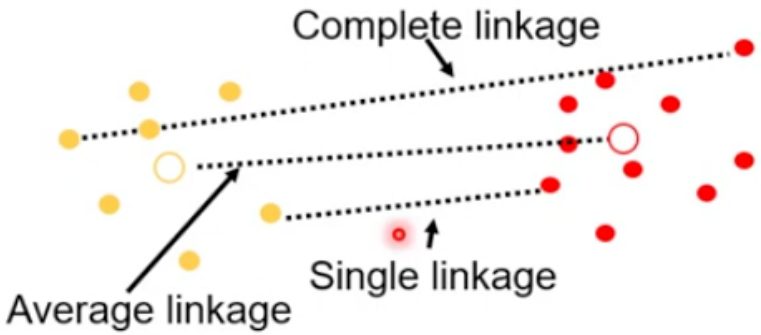
Agglomeration method

The three common distance measures between groups are:

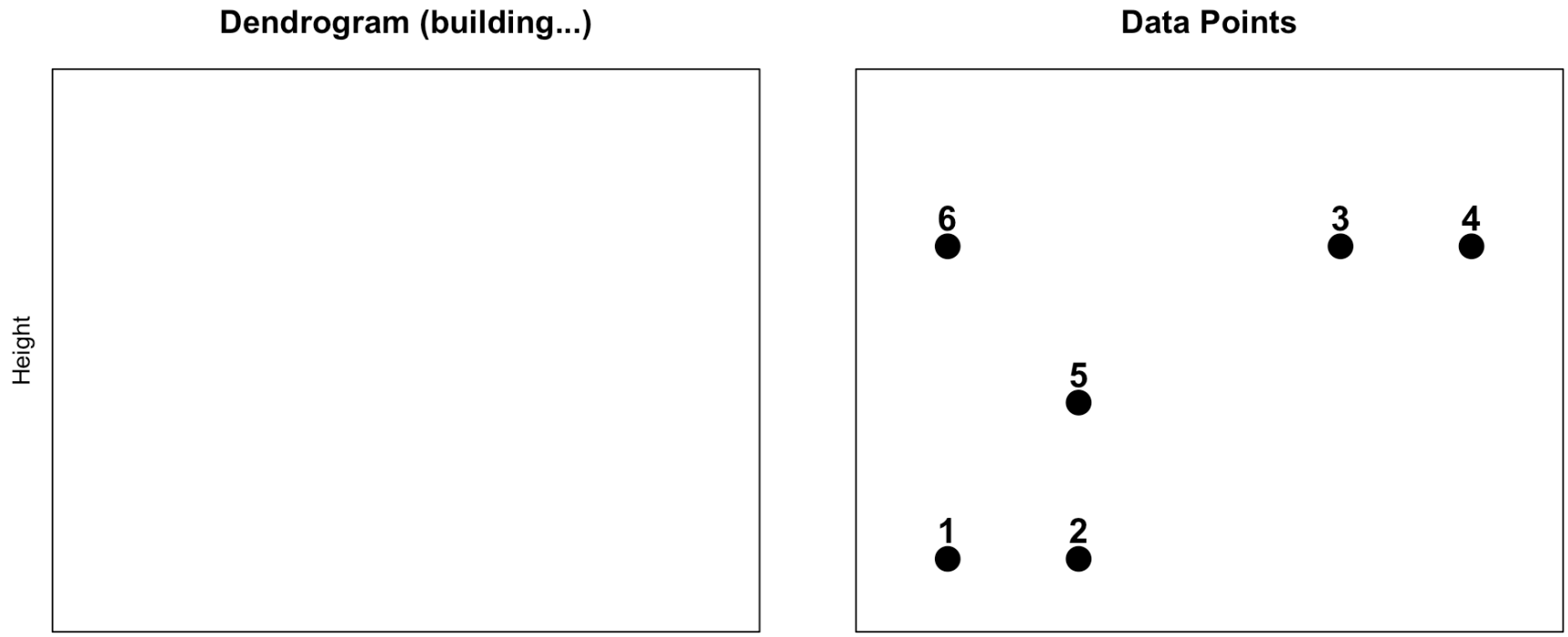
1. Single: Distance between groups = distance between closest group members

2. Average: Distance between groups = distance between group average points

3. Complete: Distance between groups = distance between most distant group members

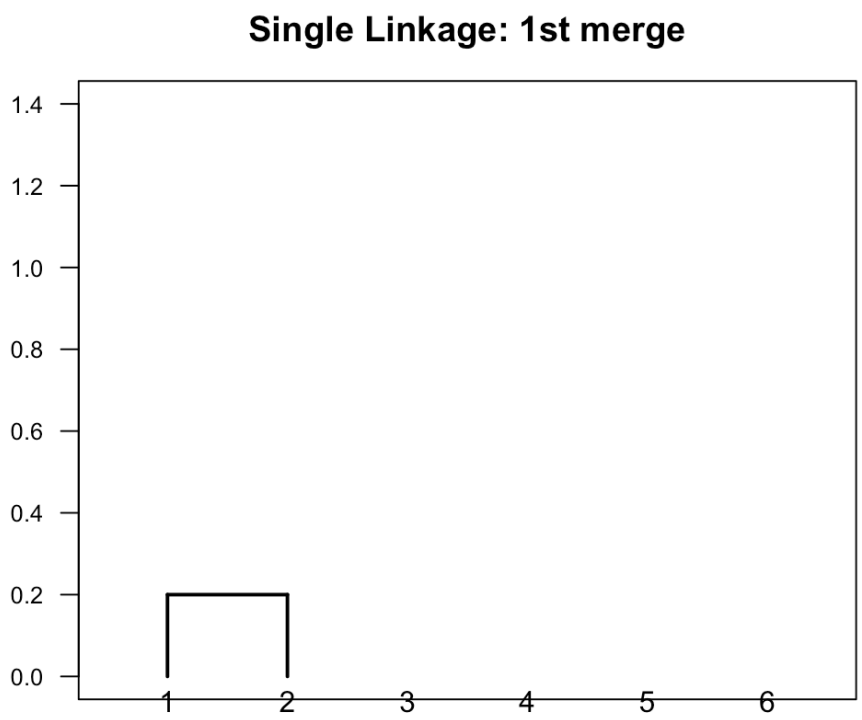


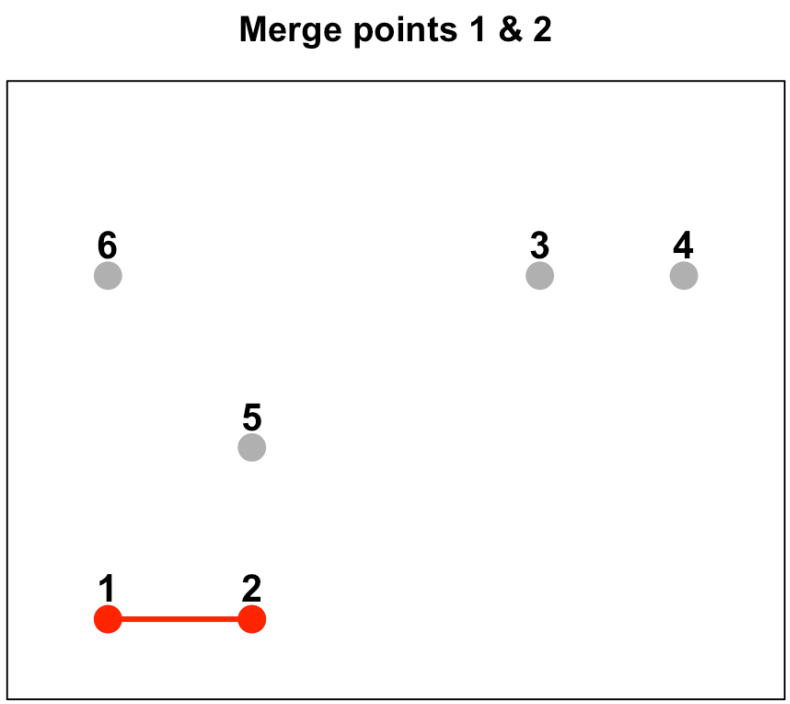
Single Linkage: Initial Points



6 clusters (each point is its own cluster)

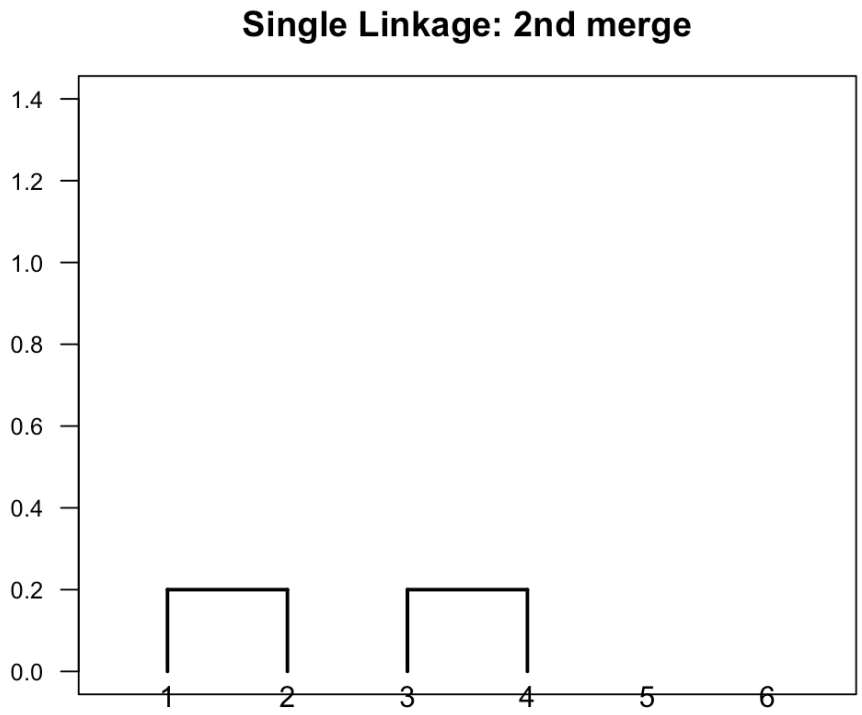
Single Linkage: Step 1 (merge 1 & 2)

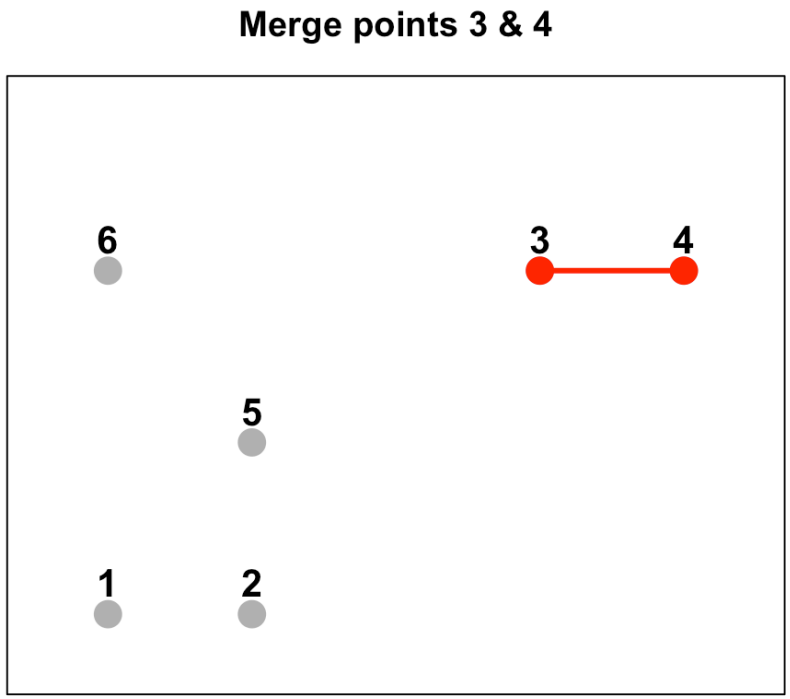




Closest pair: Points 1 & 2 (distance = 0.20) → **5 clusters**

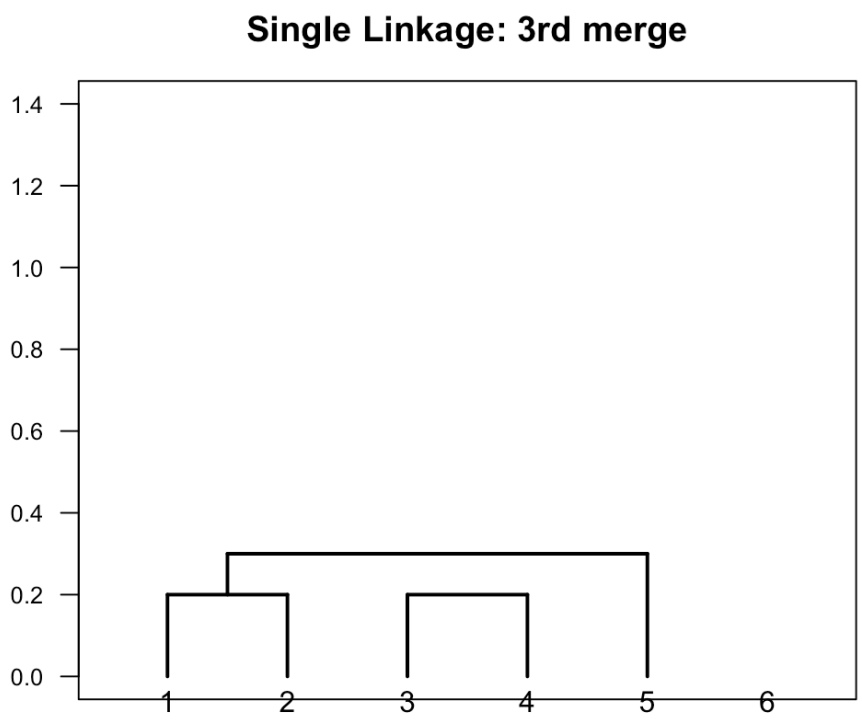
Single Linkage: Step 2 (merge 3 & 4)

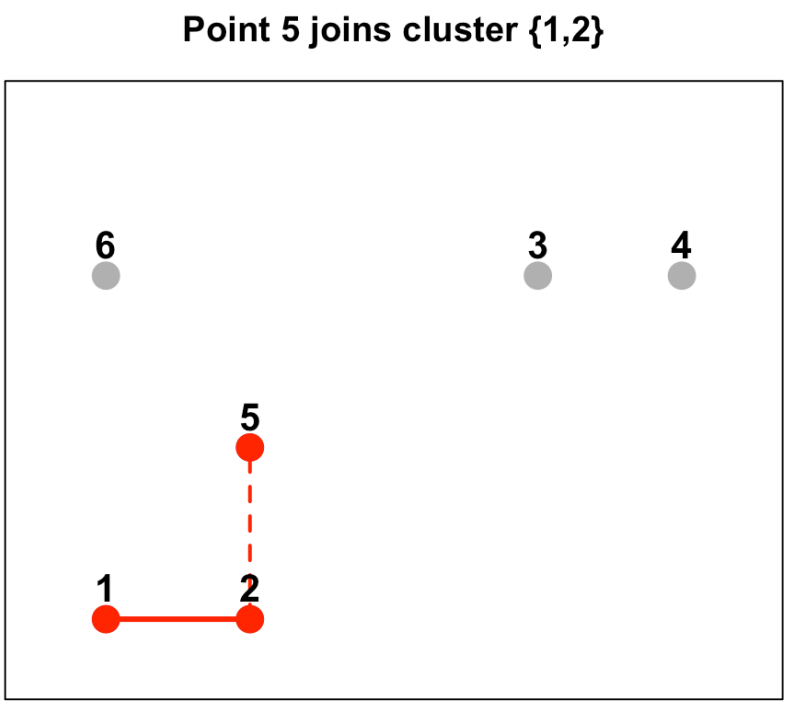




Next closest pair: Points 3 & 4 (distance = 0.20) → **4 clusters**: {1,2}, {3,4}, {5}, {6}

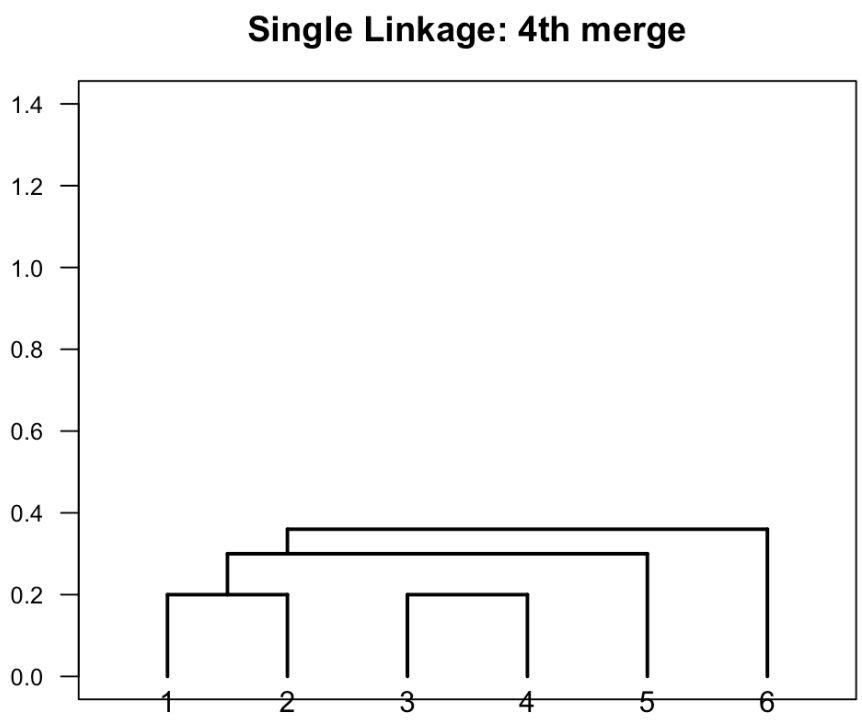
Single Linkage: Step 3 (5 joins {1,2})

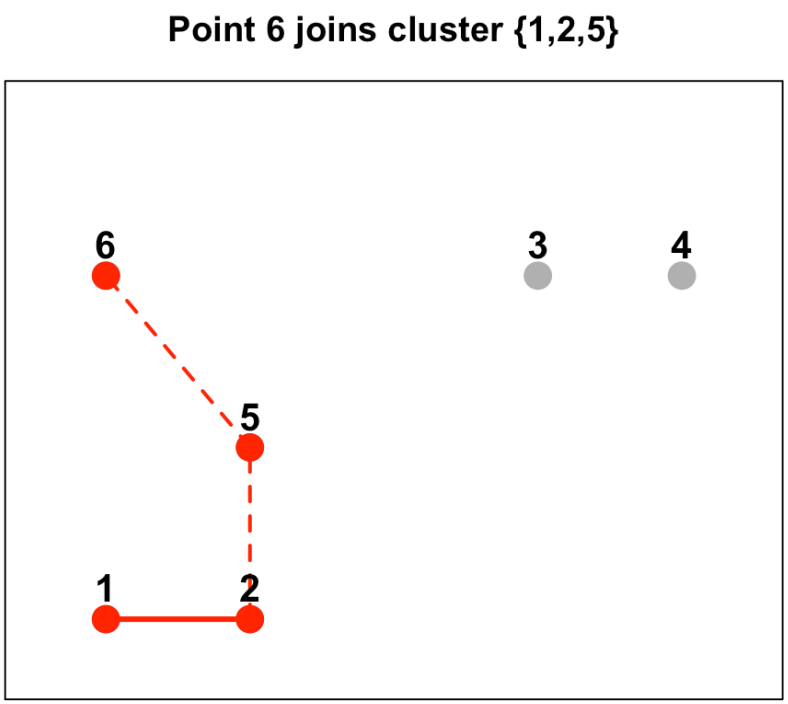




Point 5 joins {1,2} at height ≈ 0.30 → **3 clusters**: {1,2,5}, {3,4}, {6}

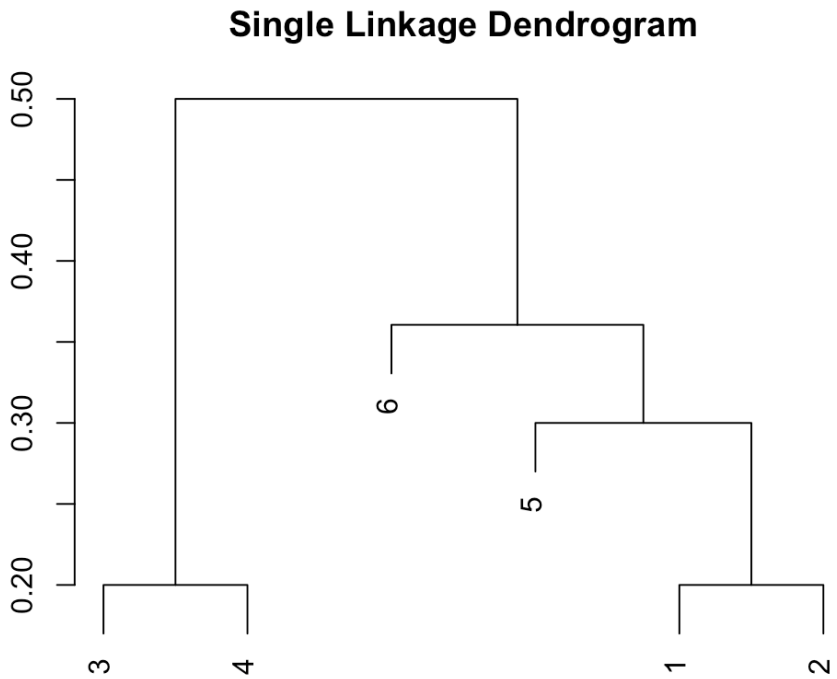
Single Linkage: Step 4 (6 joins {1,2,5})

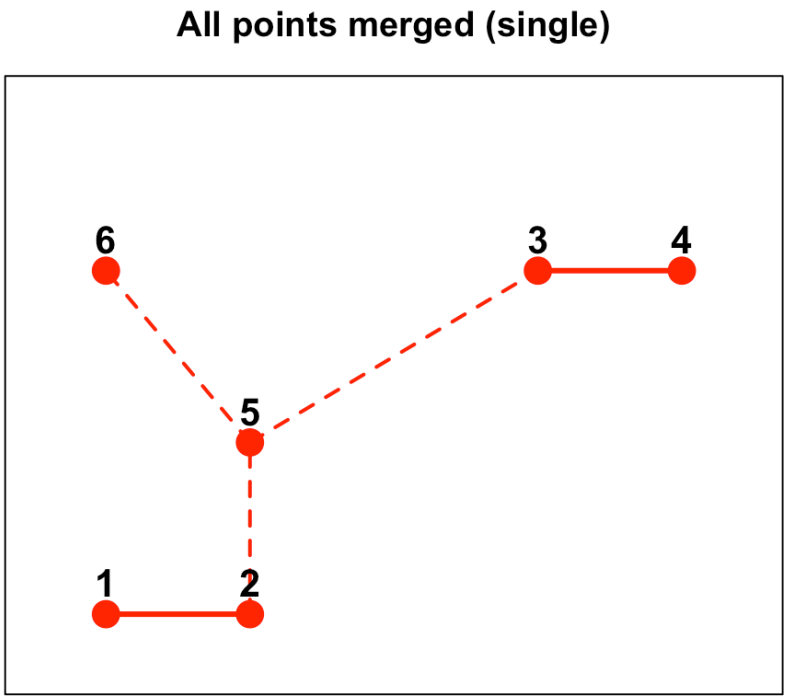




Point 6 joins {1,2,5} at height ≈ 0.36 (nearest neighbor is point 5) → **2 clusters**: {1,2,5,6}, {3,4}

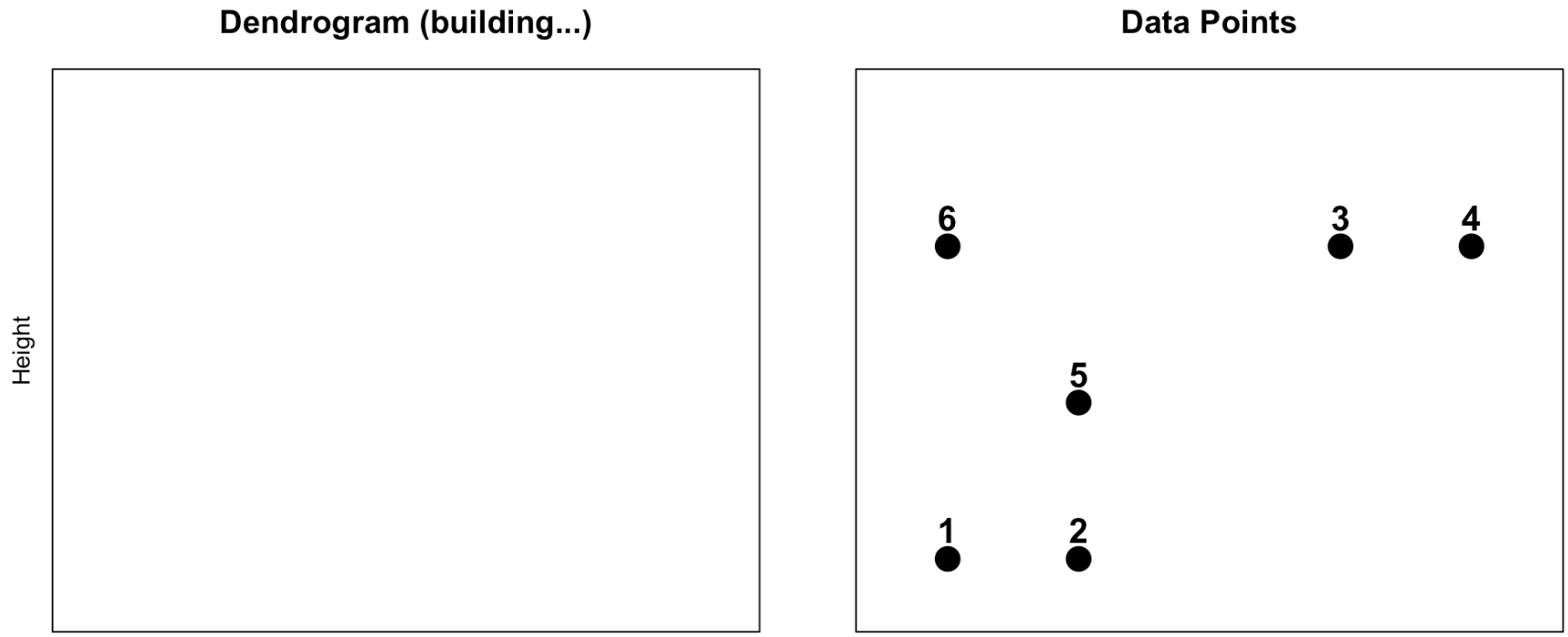
Single Linkage: Step 5 (final merge)





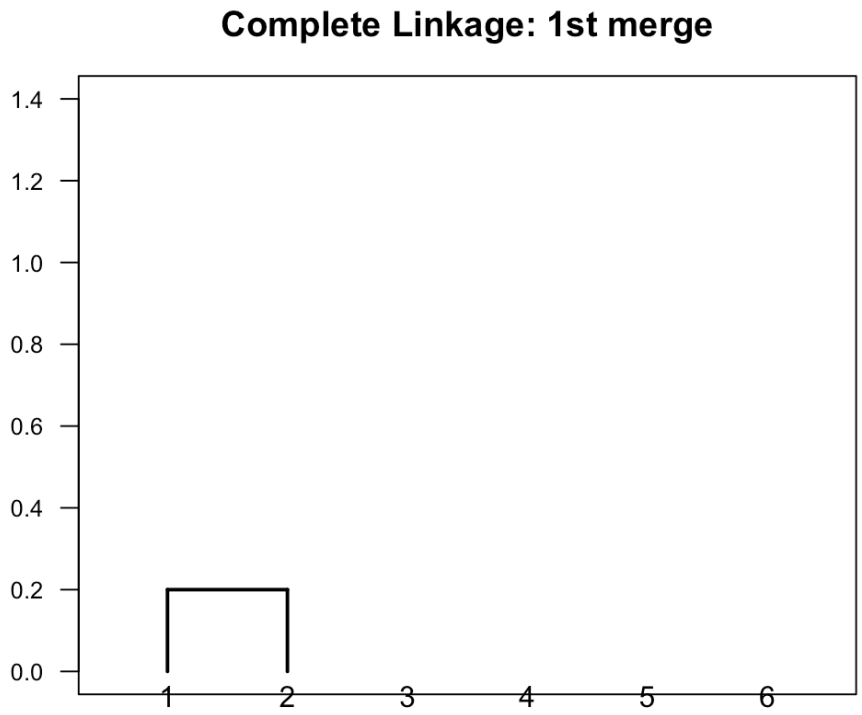
Final merge: Clusters {1,2,5,6} and {3,4} join at height ≈ 0.50 → **1 cluster**

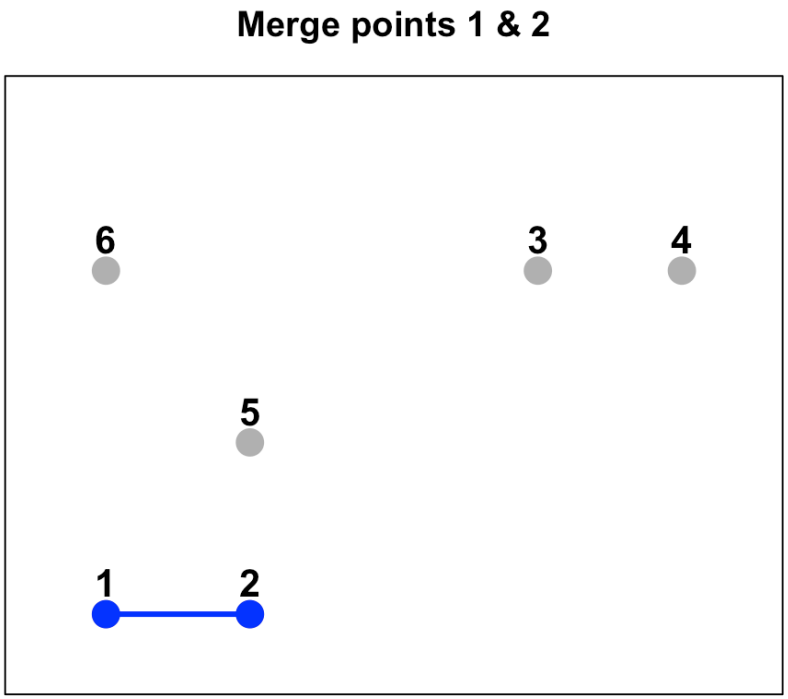
Complete Linkage: Initial Points



6 clusters (each point is its own cluster)

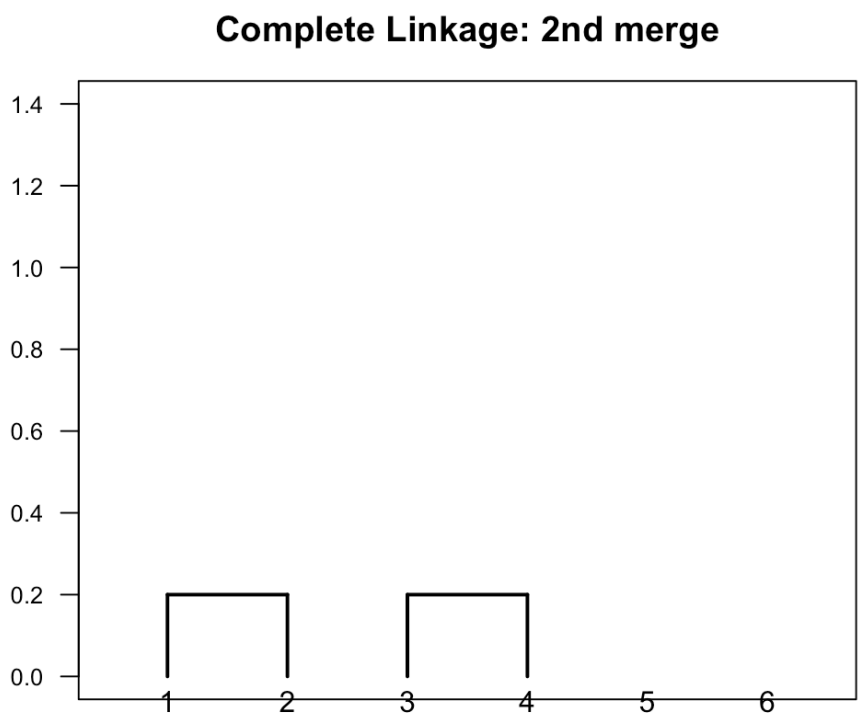
Complete Linkage: Step 1 (merge 1 & 2)

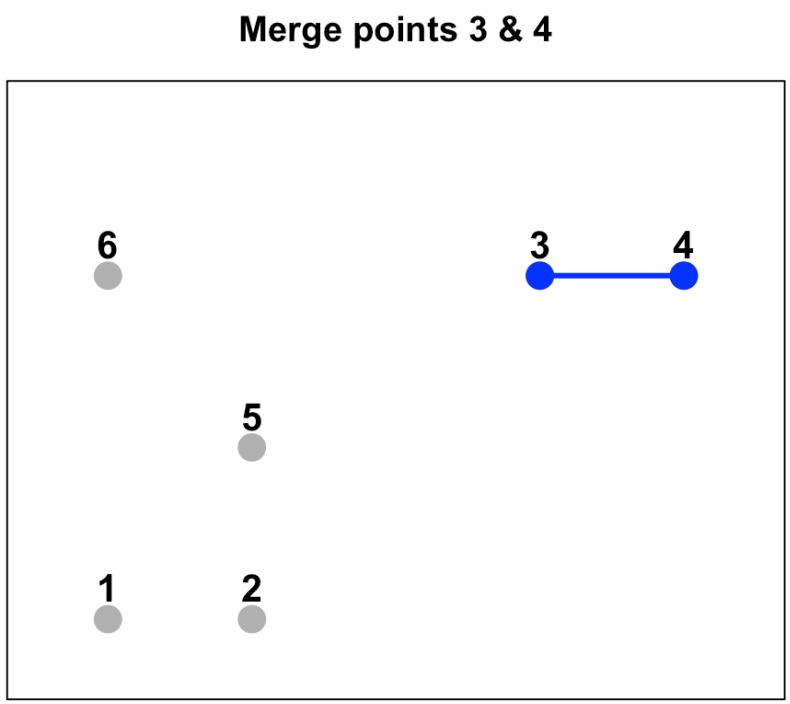




Closest pair: Points 1 & 2 (distance = 0.20) → **5 clusters**

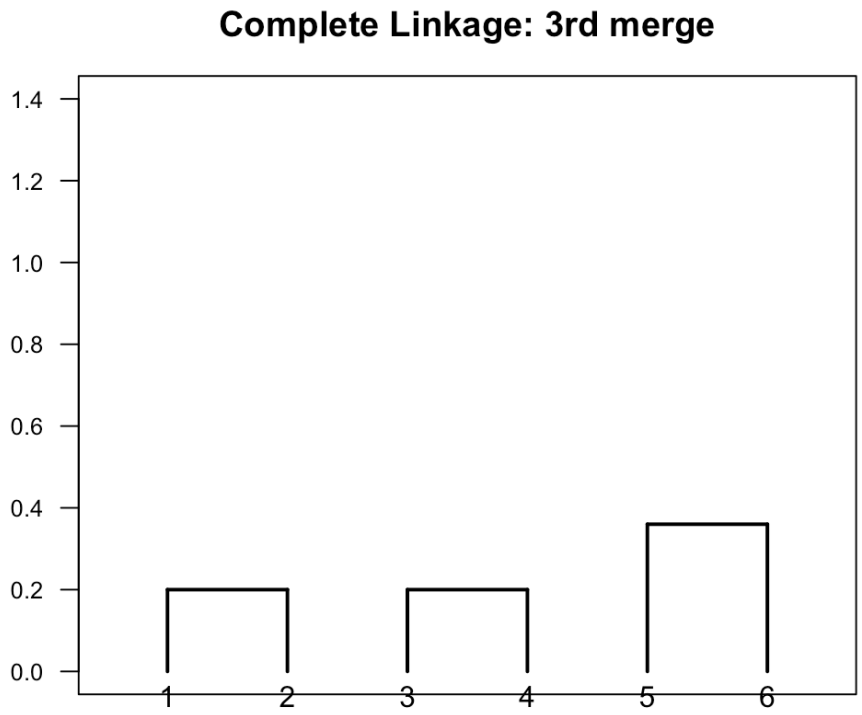
Complete Linkage: Step 2 (merge 3 & 4)

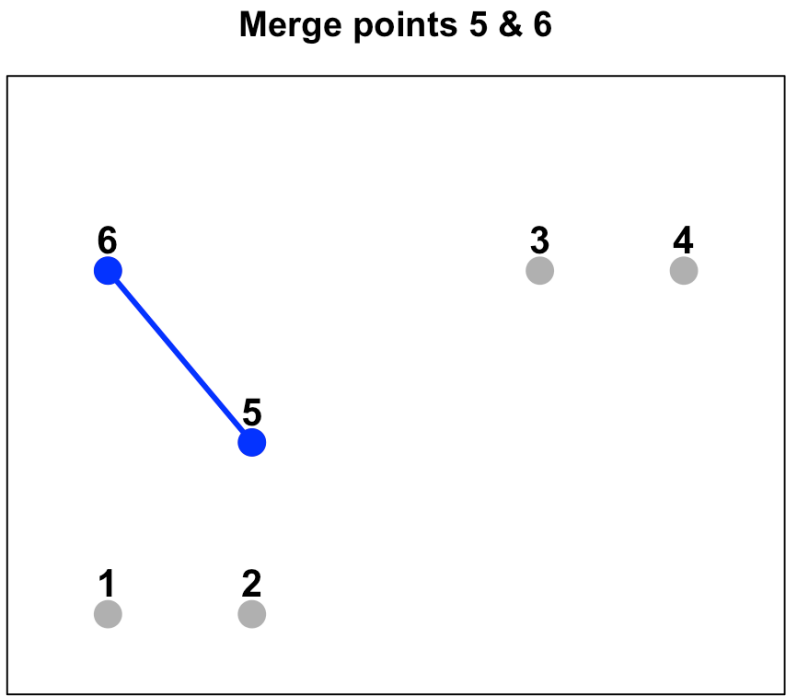




Next closest pair: Points 3 & 4 (distance = 0.20) → **4 clusters**: {1,2}, {3,4}, {5}, {6}

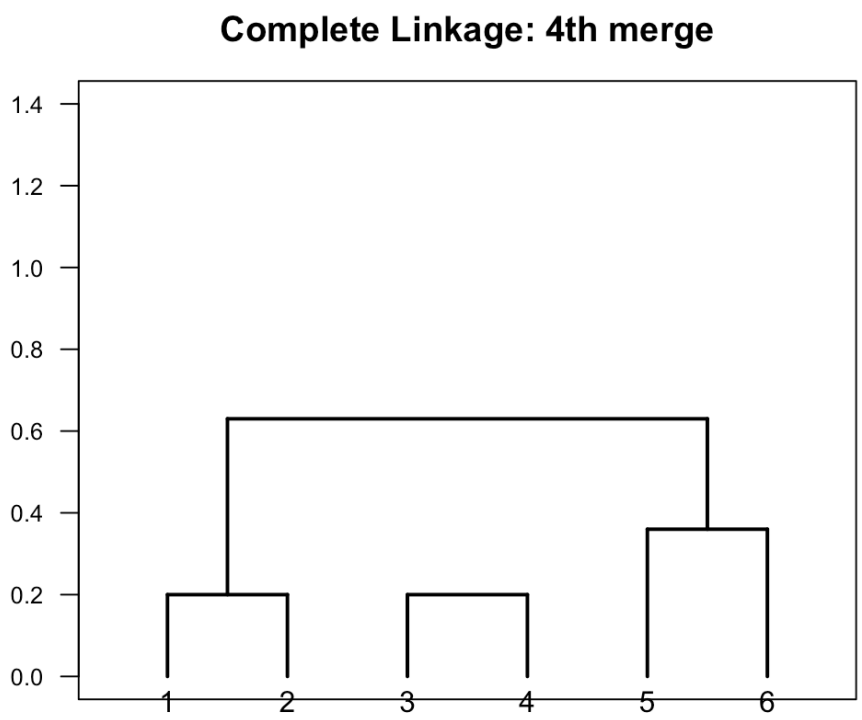
Complete Linkage: Step 3 (merge 5 & 6)

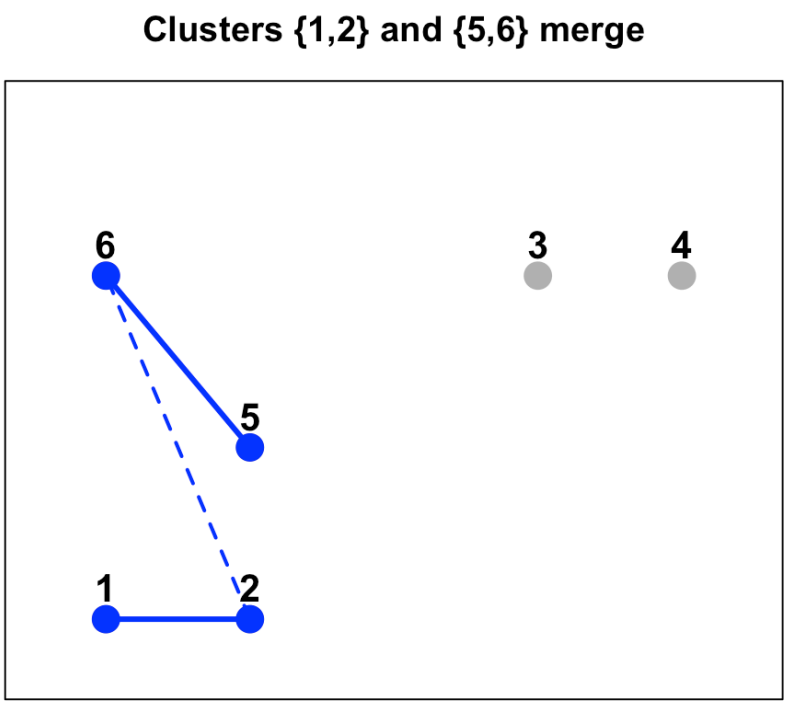




Points 5 & 6 form a cluster at height ≈ 0.36 → **3 clusters**: {1,2}, {3,4}, {5,6}

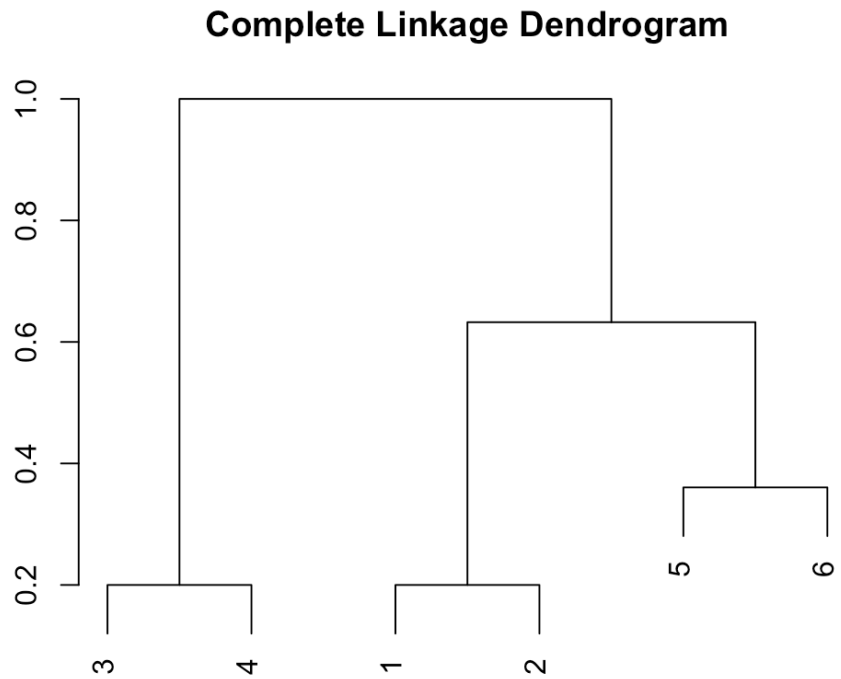
Complete Linkage: Step 4 ({1,2} joins {5,6})

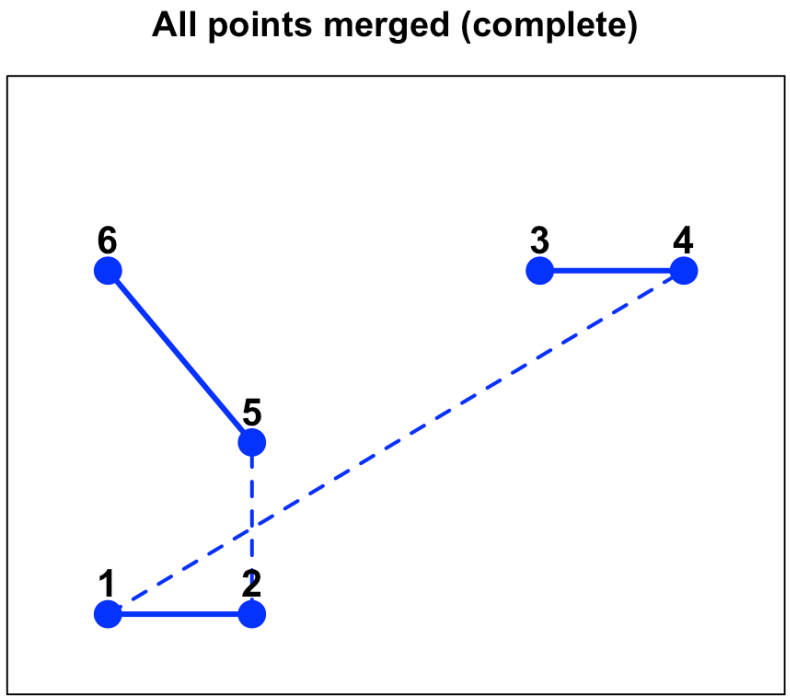




Clusters {1,2} and {5,6} merge at height equal to the maximum distance between any point in {1,2} and any point in {5,6}, which in this case is the distance between points 6 and 2 (≈ 0.63) → **2 clusters**: {1,2,5,6}, {3,4}

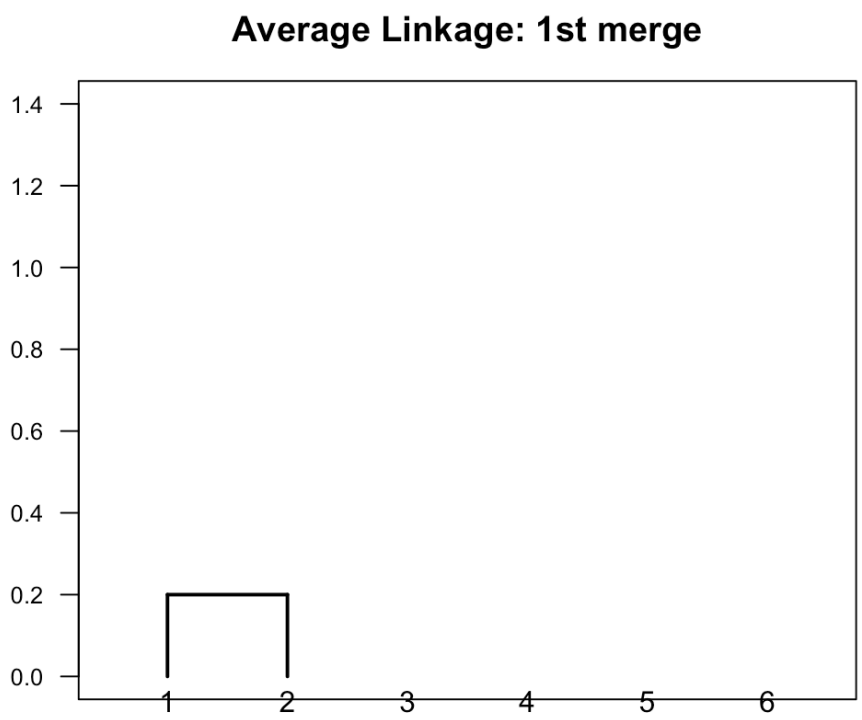
Complete Linkage: Step 5 (final merge)

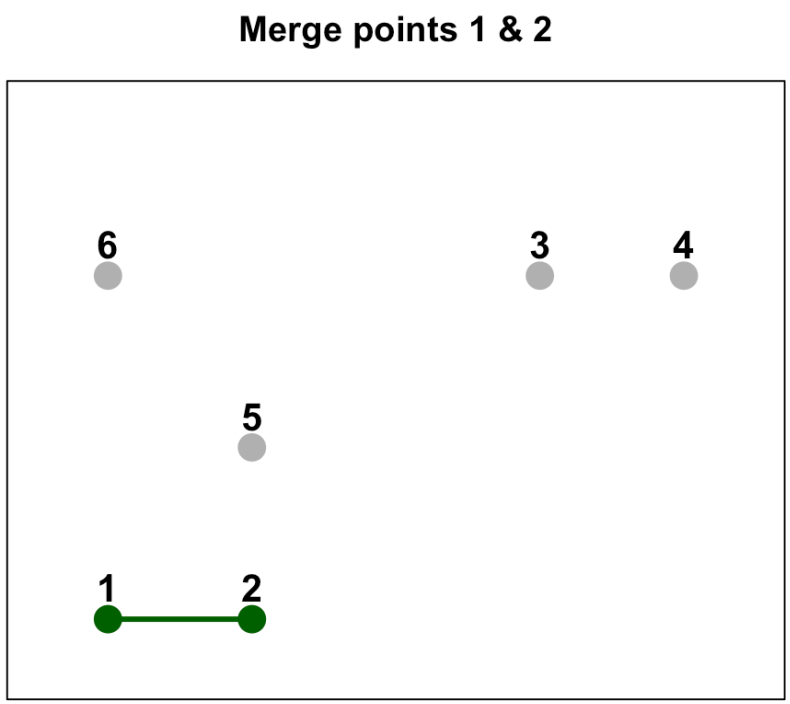




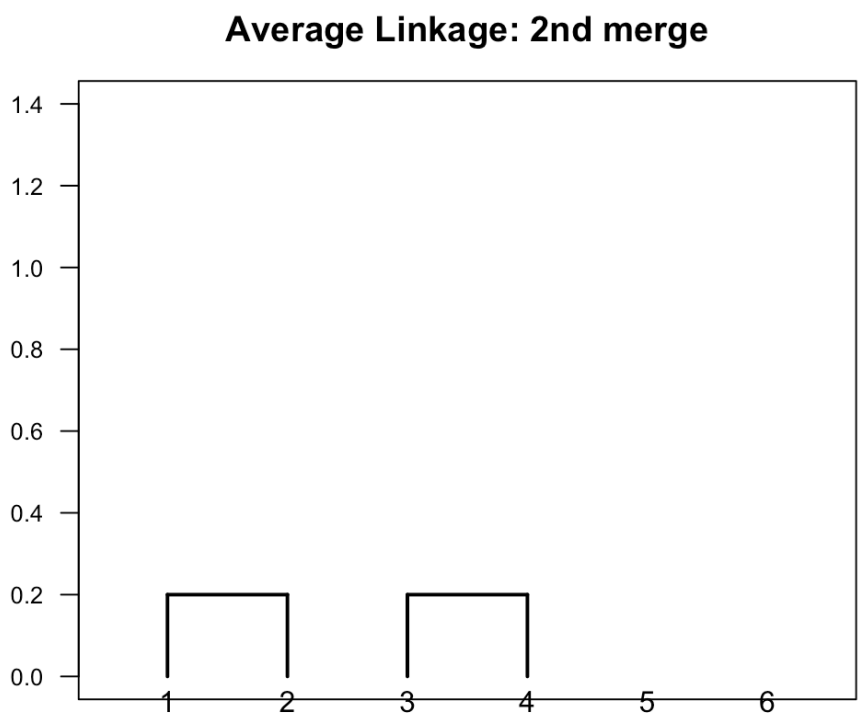
Final merge: Clusters {1,2,5,6} and {3,4} join at height ≈ 1.00 → **1 cluster**

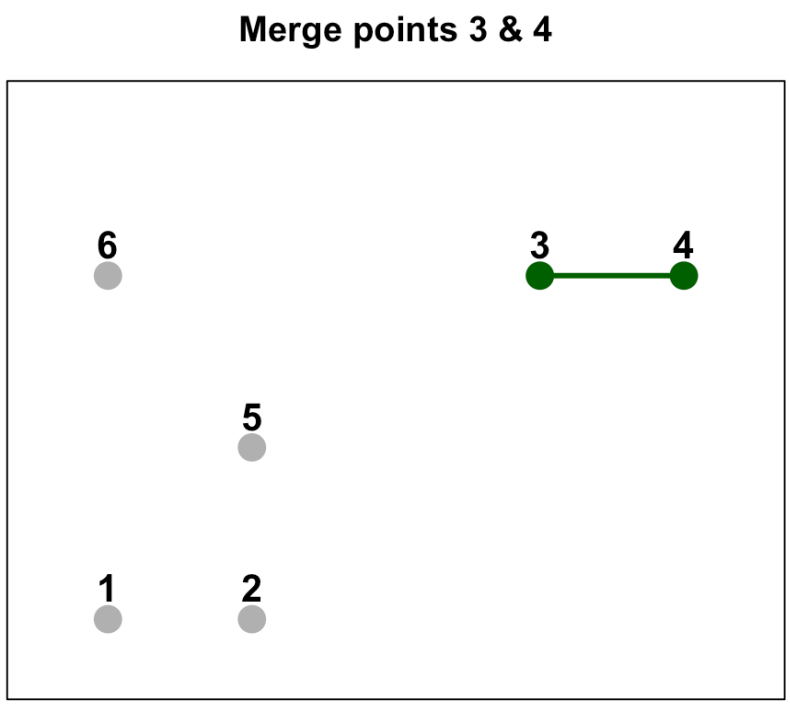
Average Linkage: Step 1 (merge 1 & 2)



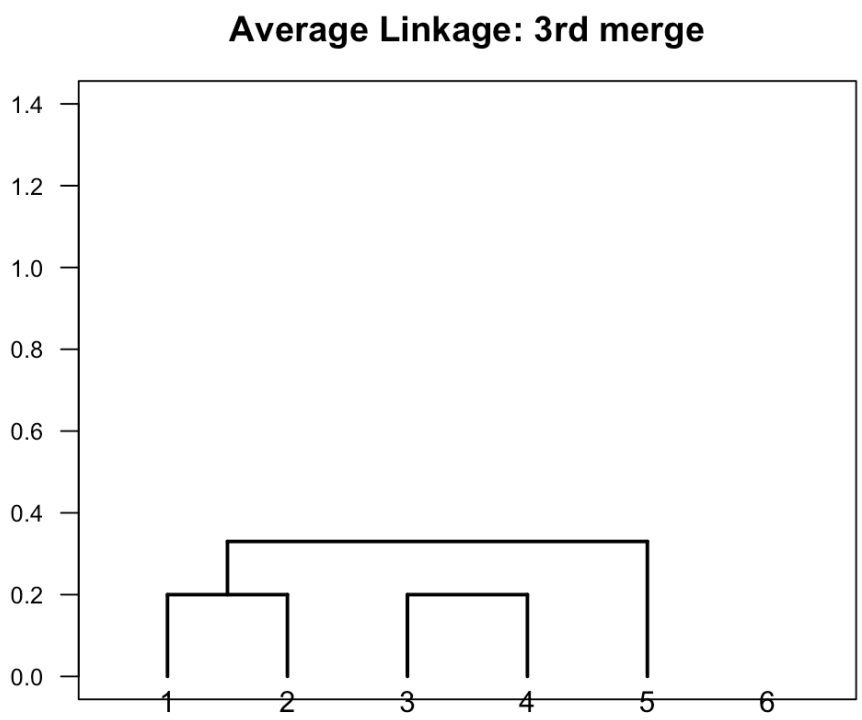


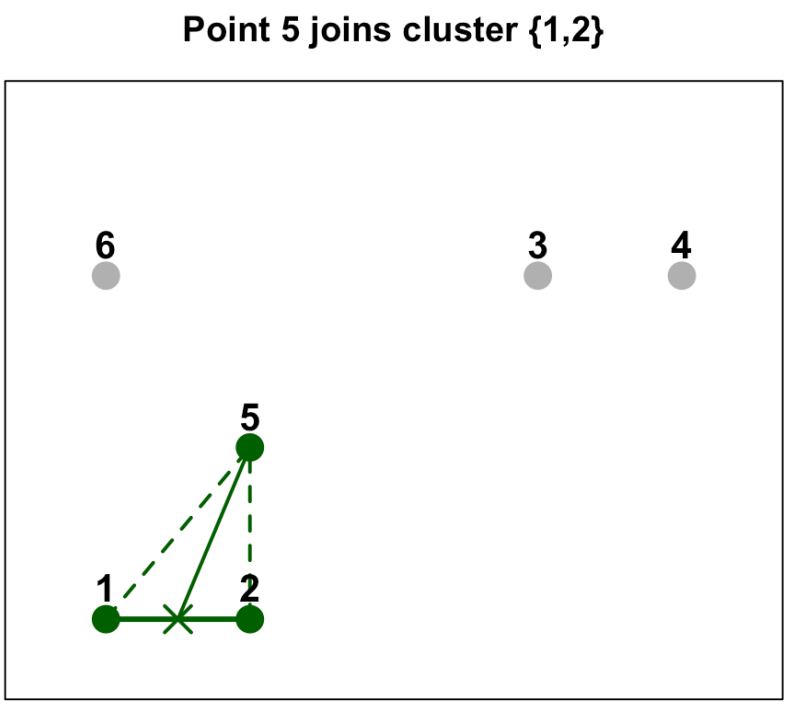
Average Linkage: Step 2 (merge 3 & 4)



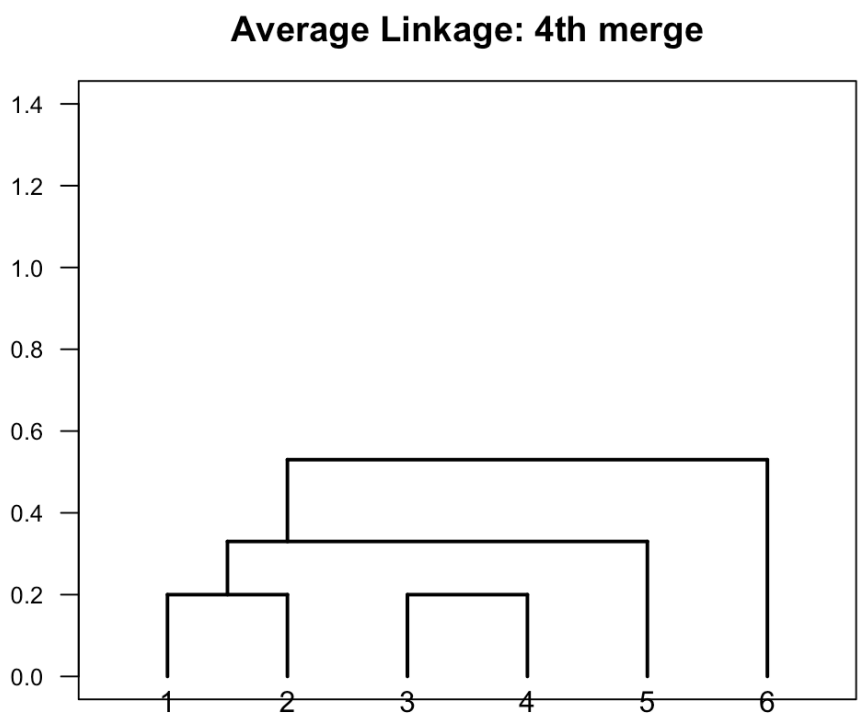


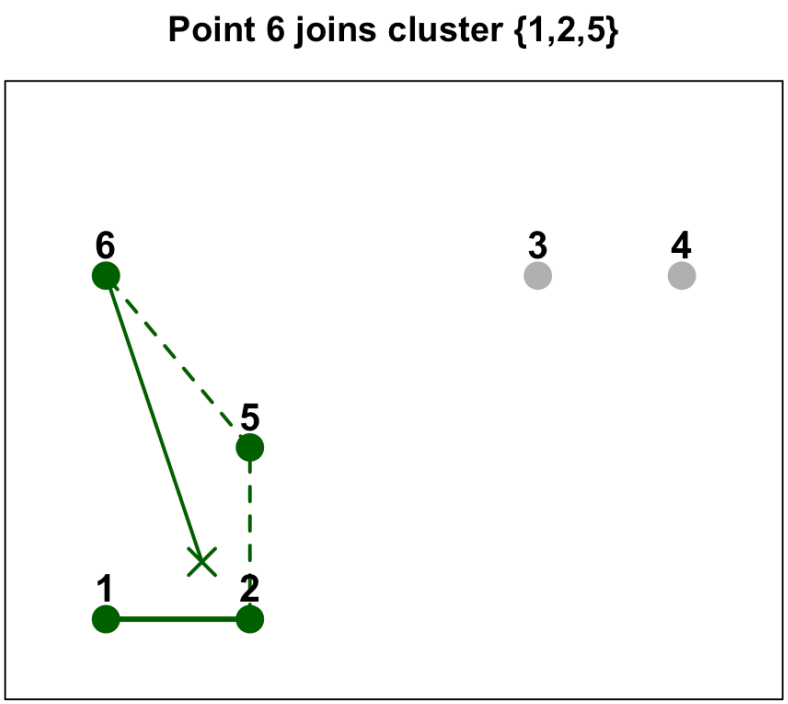
Average Linkage: Step 3 (5 joins {1,2})



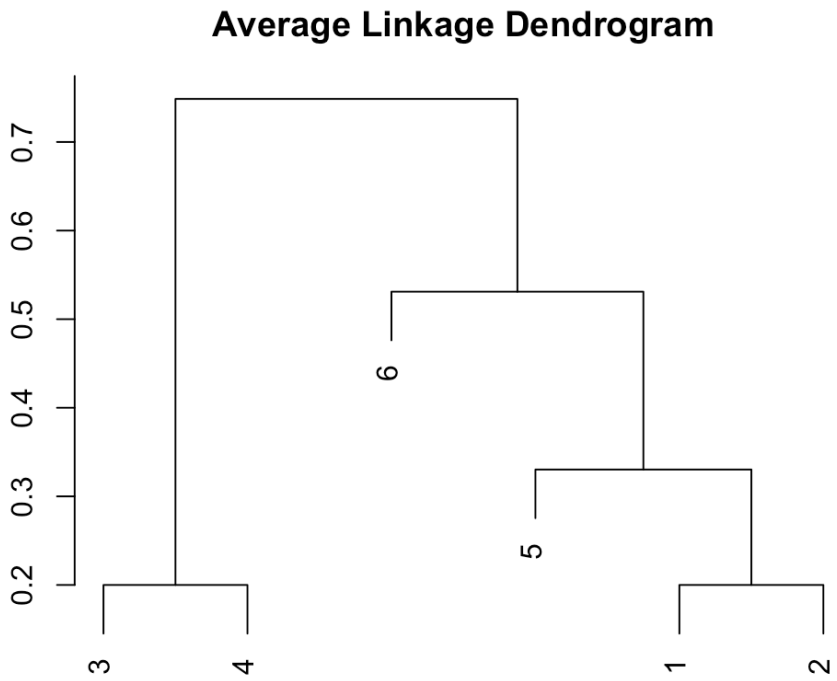


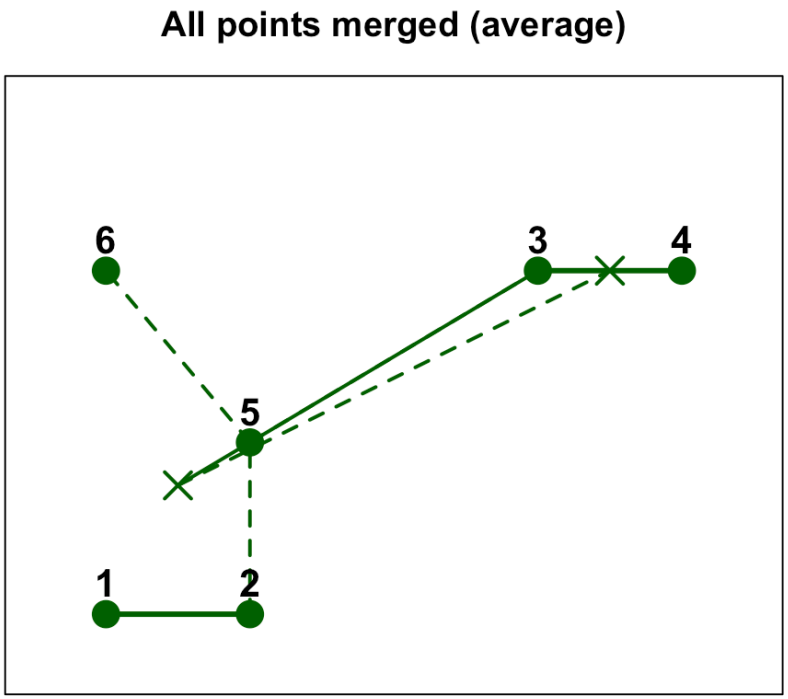
Average Linkage: Step 4 (6 joins {1,2,5})





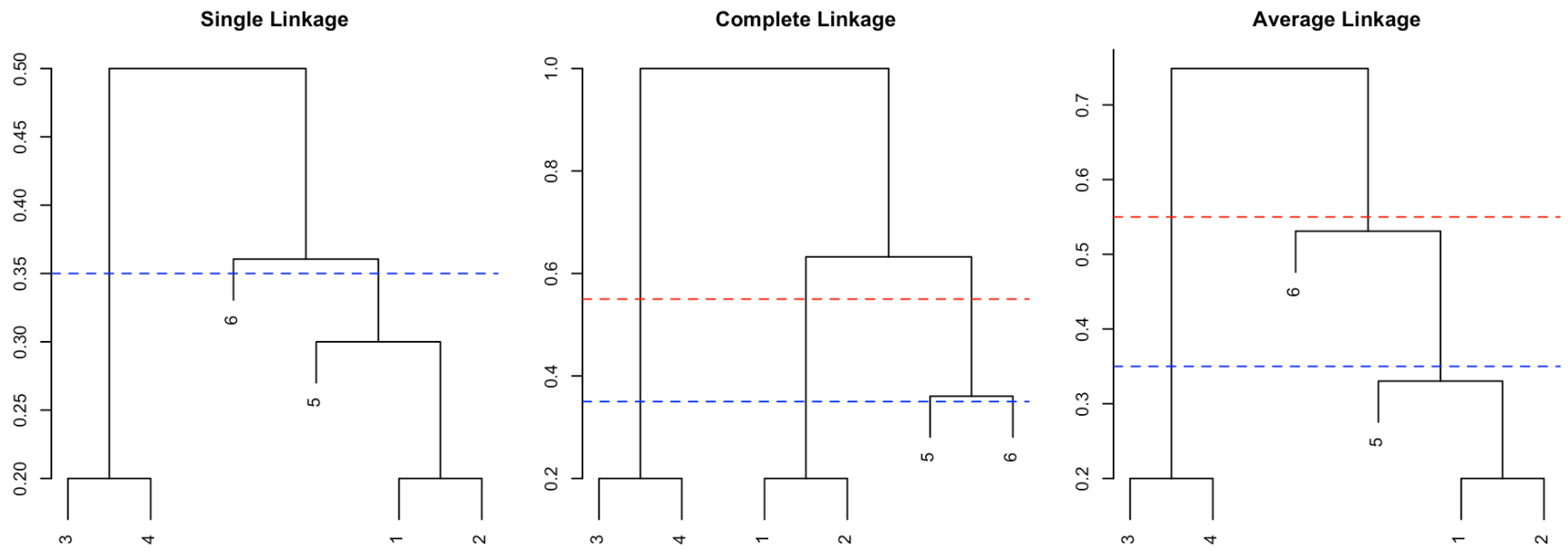
Average Linkage: Step 5 (final merge)





Final merge: Clusters {1,2,5,6} and {3,4} join at height ≈ 0.75 → **1 cluster**

Comparison: Different Linkage, Different Clusters

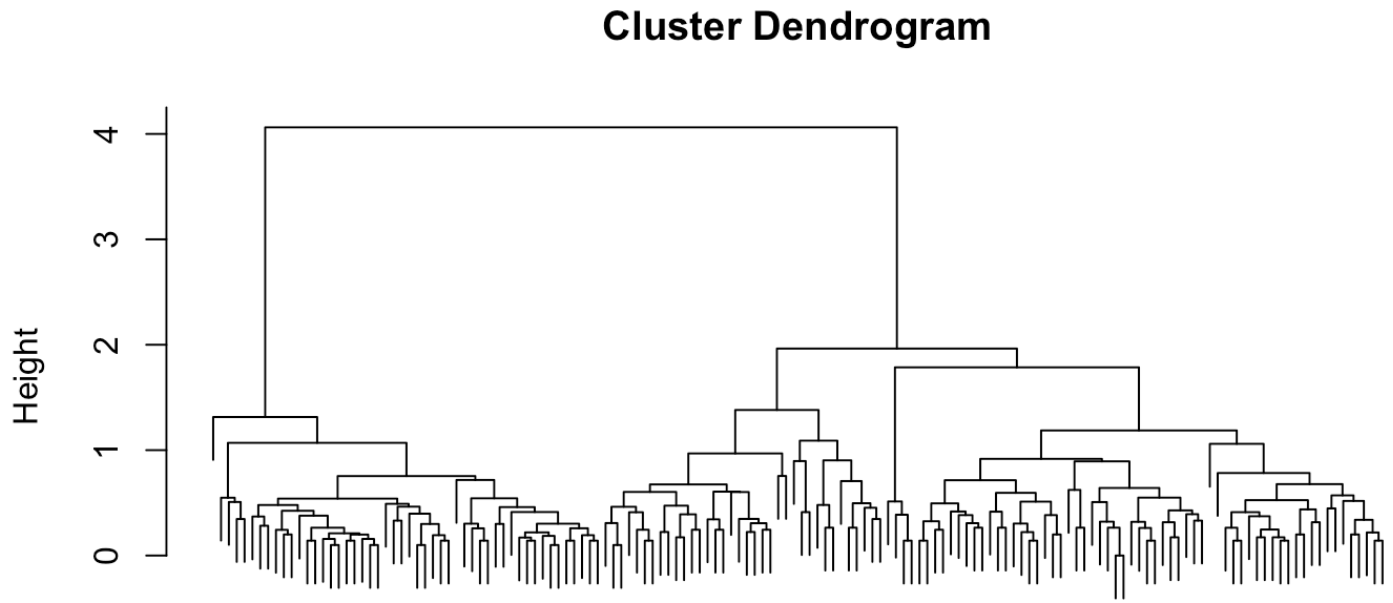


So the same data and same cut height give **different clusterings** for different linkage methods.

Hierarchical clustering in R

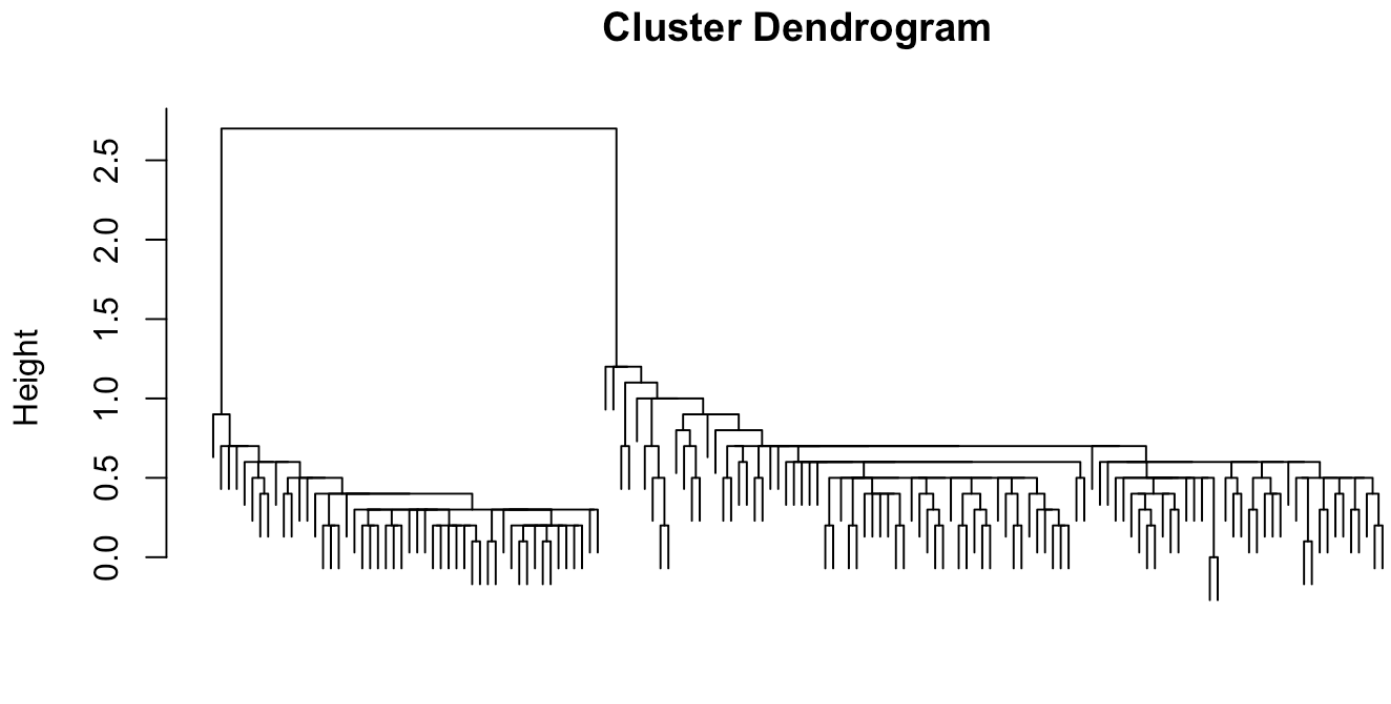
In Rstudio we can use the `hclust()-function` to perform hierarchical clustering.

```
dd <- dist(df_iris, method="euclidean")
hc <- hclust(dd, method="average")
plot(hc, labels = FALSE)
```



Example using Euclidean distance and single linkage

```
dd <- dist(df_iris, method="manhattan")
hc <- hclust(dd, method="single")
plot(hc, labels = FALSE)
```



Choosing the number of clusters

With a dendrogram, we **cut** it at some height to obtain K clusters `cutree(hc, k = 3)` or `cutree(hc, h = 3.5)`.

How to choose the cut height or K?

1. **Visual inspection**: Look for large jumps in height — cut just before a big jump.
2. **Domain knowledge**: If you expect K groups based on the problem context.
3. **Internal validation**: Use quality metrics (e.g., **silhouette** coefficient, **within-cluster sum of squares, F scores**) for different K.
4. **External validation (if labels available)**: Compare at different heights to ground-truth using the **Adjusted Rand Index (ARI)**. Range [−1, 1]; 1 = perfect, 0 = random.

Compare hc and km clusters K = 3

Compare hc and km clusters K = 3

```
adjustedRandIndex(cutree(hc, k = 3), iris$Species)
```

```
## [1] 0.7591987
```

```
adjustedRandIndex(km$cluster, iris$Species)
```

```
## [1] 0.7302383
```

```
wcss(cutree(hc, k = 3))
```

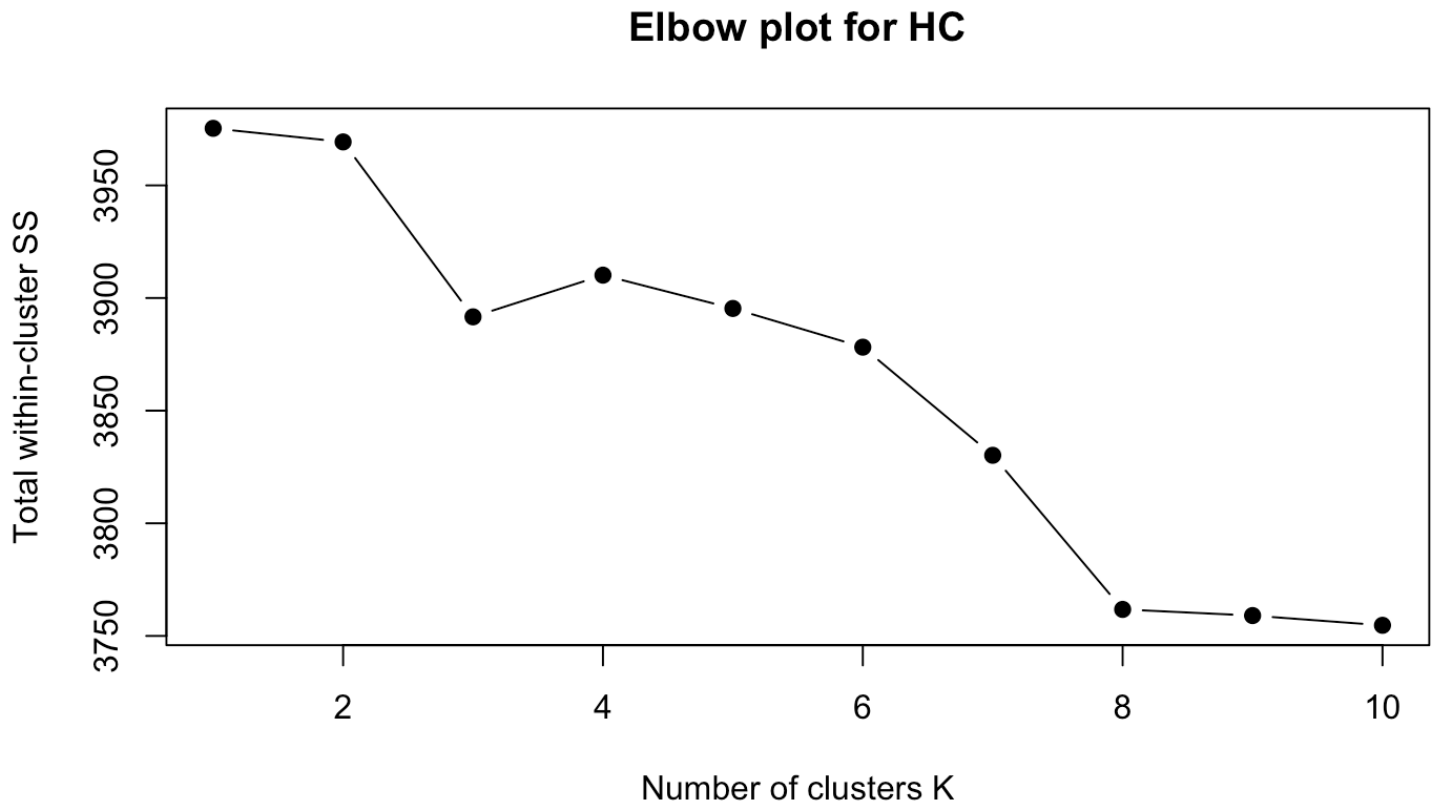
```
## [1] 3891.643
```

```
wcss(km$cluster)
```

```
## [1] 3924.729
```

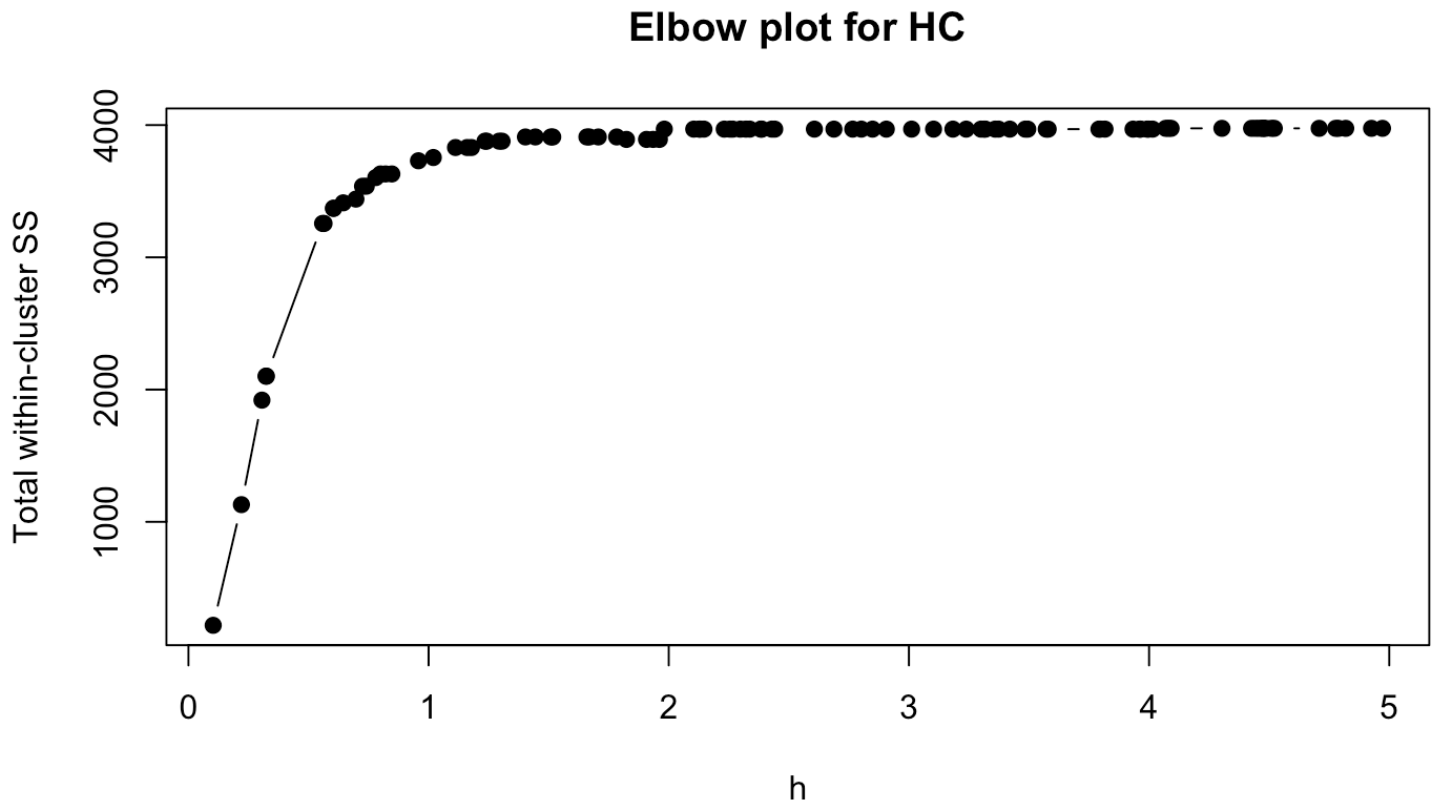
Elbow plots for WCSS and K

Do not need to recompute! Same dendrogram.



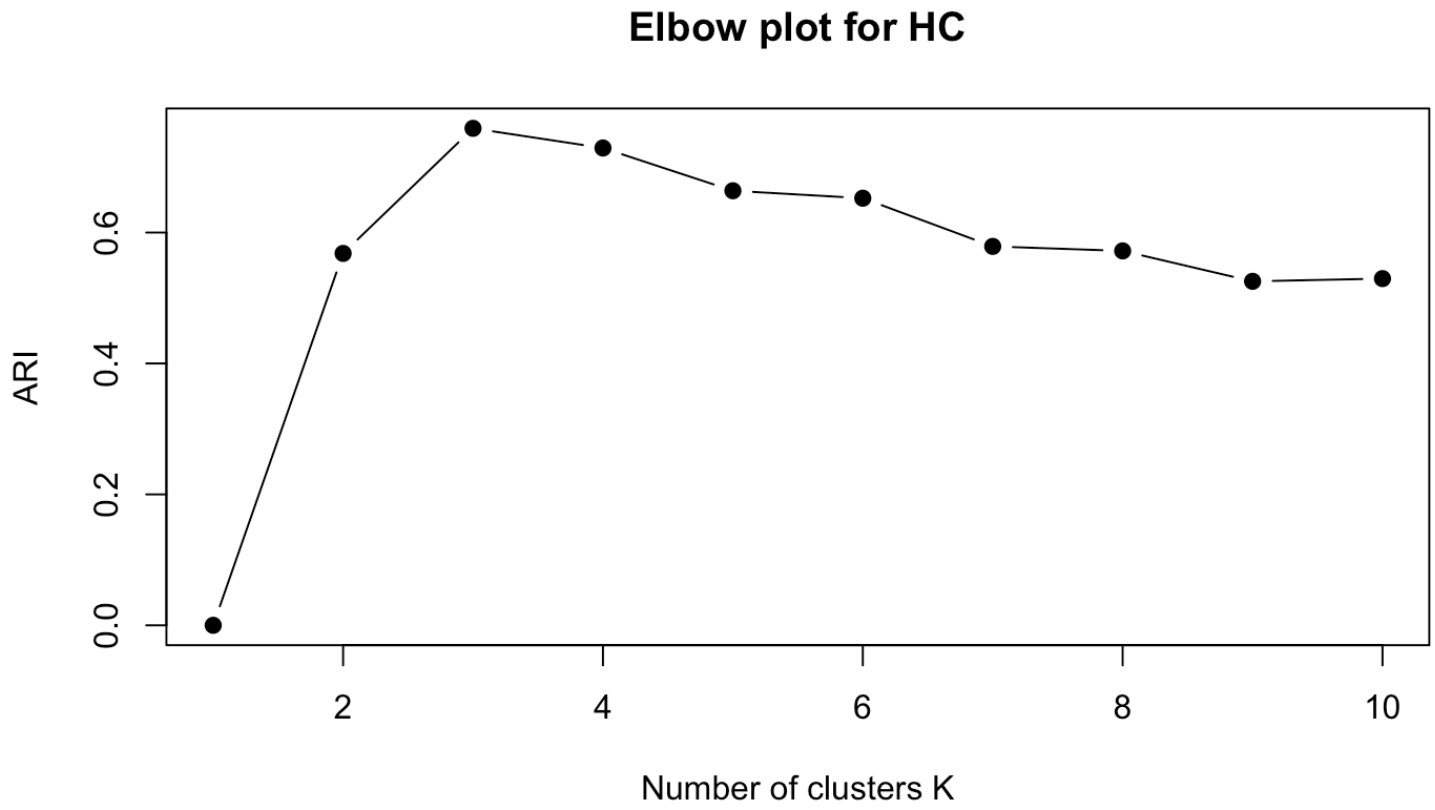
Elbow plots for WCSS and height

Do not need to recompute! Same dendrogram.



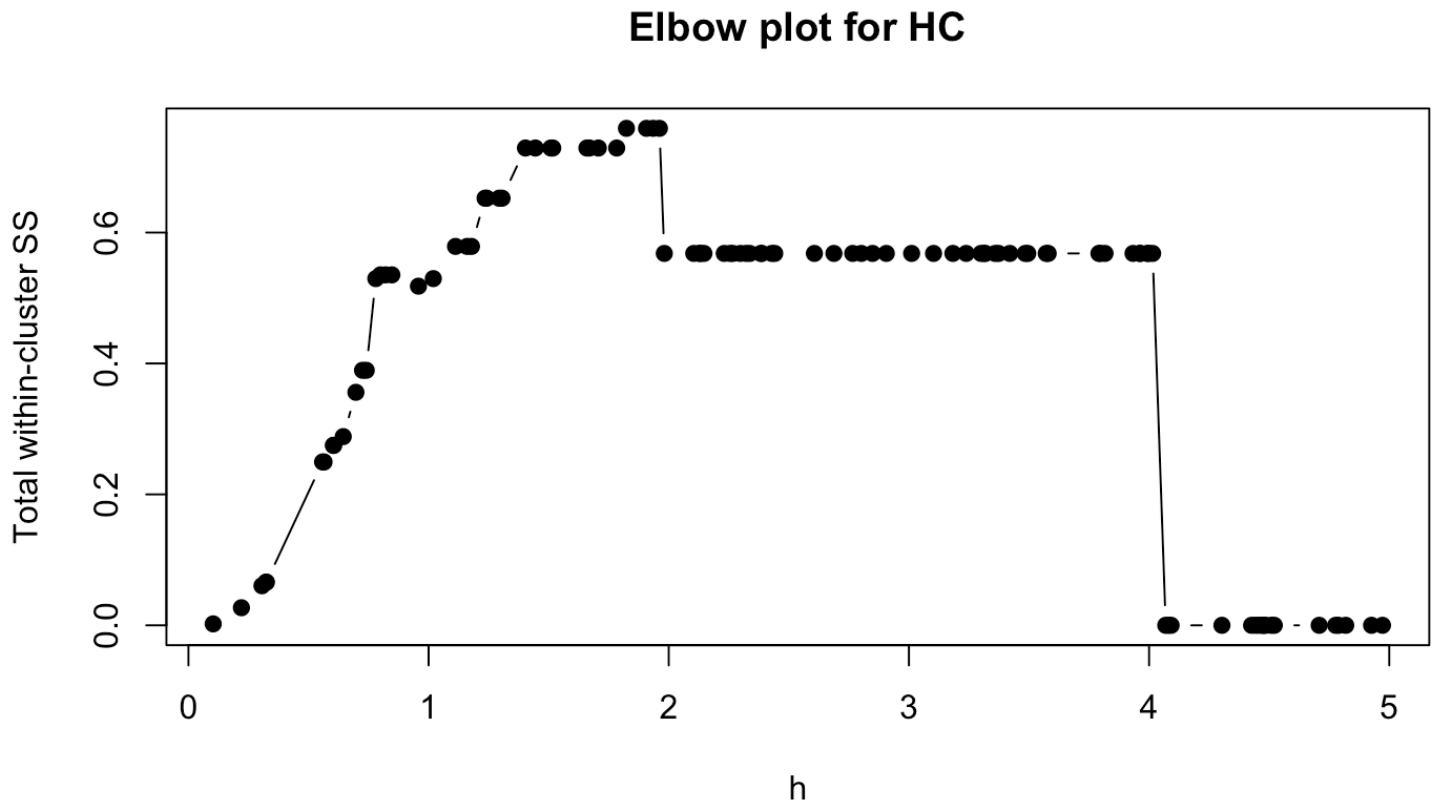
Elbow plots for ARI and K

Do not need to recompute! Same dendrogram.



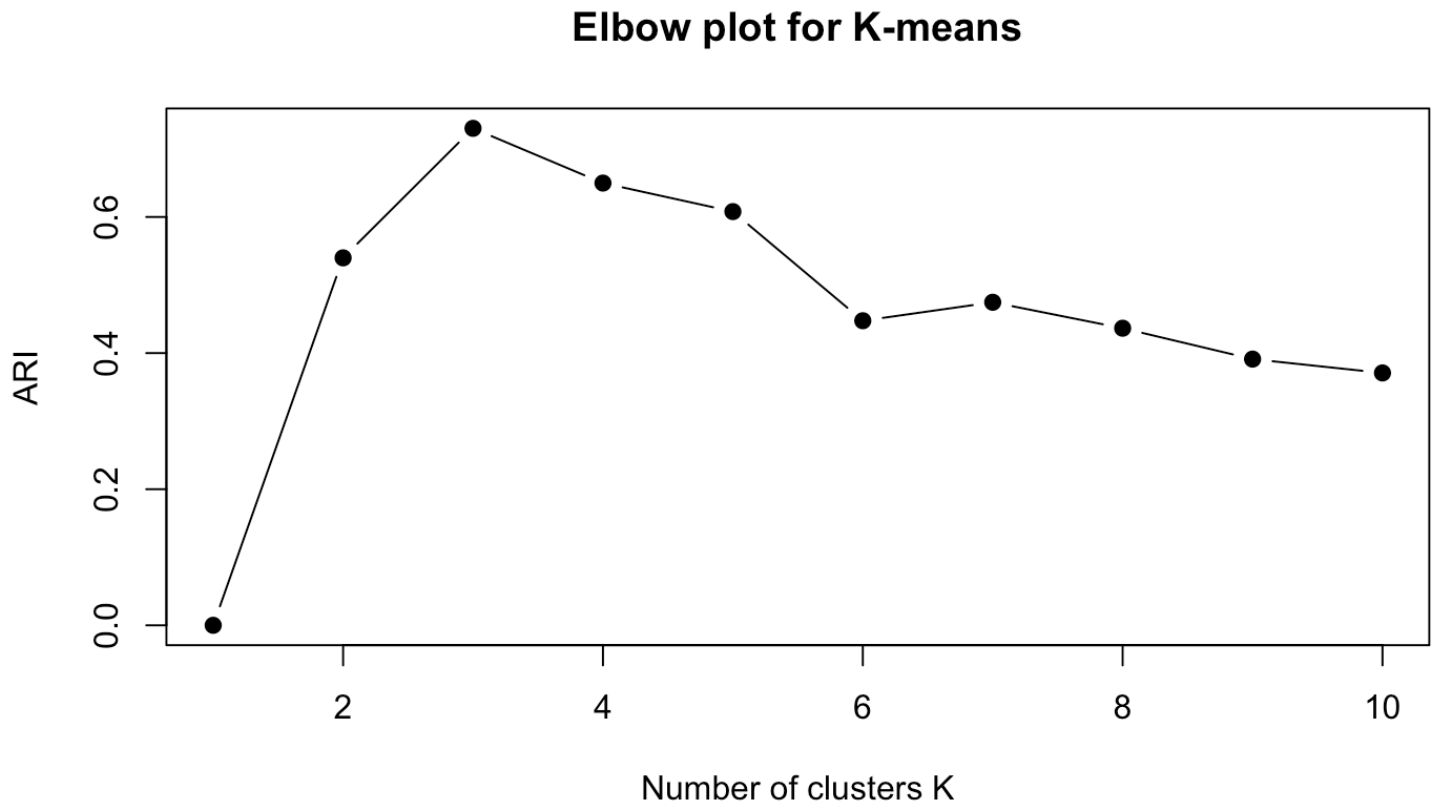
Elbow plots for ARI and height

Do not need to recompute! Same dendrogram.



Elbow plots for ARI and K for K-means

Do not need to recompute! Same dendrogram.



Silhouette coefficient

Internal validation (no ground-truth labels):

- **Silhouette coefficient** for observation i:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

  where $a(i)$ = average distance to points in same cluster,
  $b(i)$ = average distance to points in nearest other cluster.
  Range: [−1, 1]. Values near 1 = well-clustered, near 0 = borderline, negative = misclassified.

- **Average silhouette**: mean of $s(i)$ over all observations (higher is better).

Evaluating clustering quality with silhouette

```
sil <- silhouette(cutree(hc, k = 3), dist(iris[,1:4]))
mean(sil[, 3]) # Average silhouette width
```

```
## [1] 0.5541609
```

```
sil <- silhouette(cutree(hc, h = 3), dist(iris[,1:4]))
mean(sil[, 3]) # Average silhouette width
```

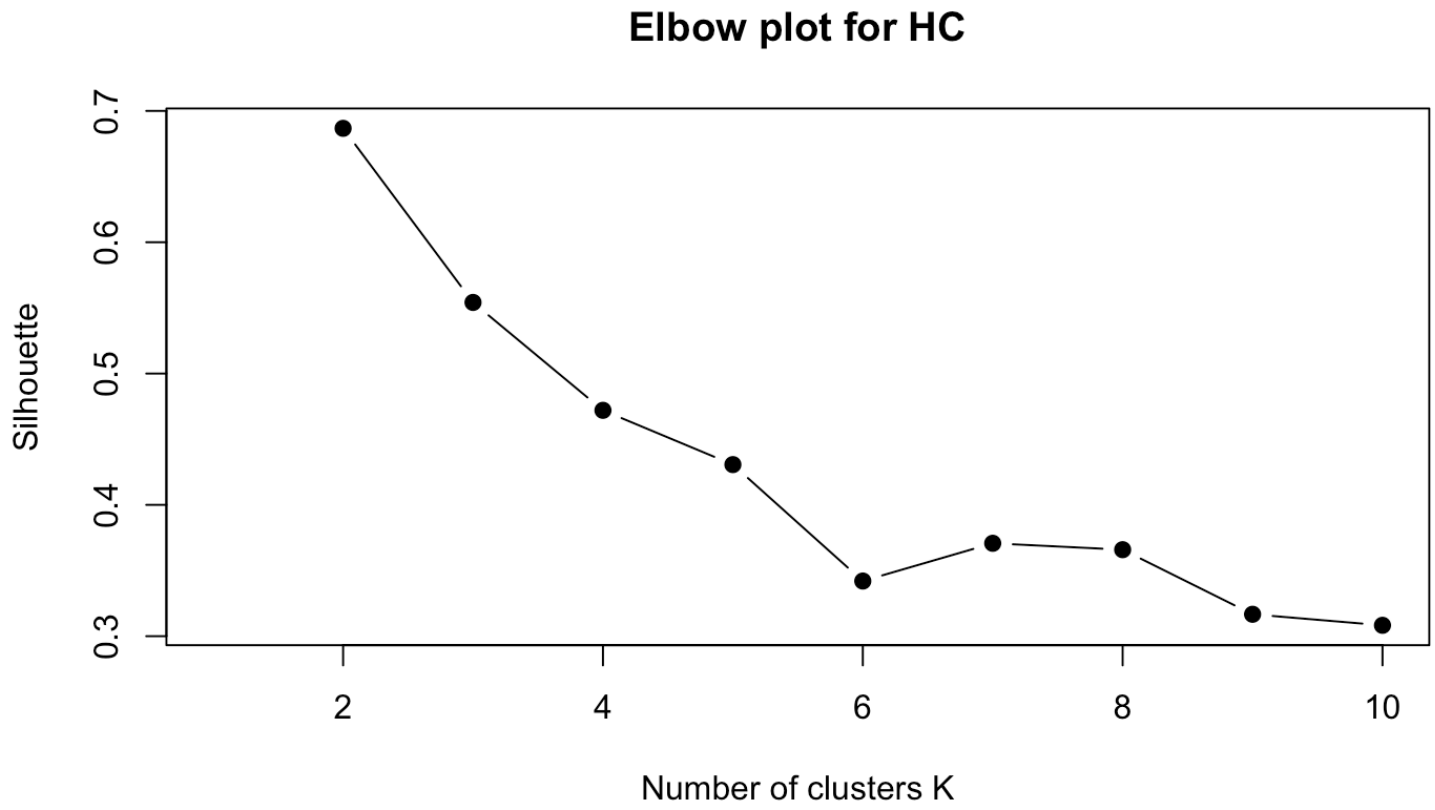
```
## [1] 0.6867351
```

```
sil <- silhouette(km$cluster, dist(iris[,1:4]))
mean(sil[, 3]) # Average silhouette width
```

```
## [1] 0.3223781
```

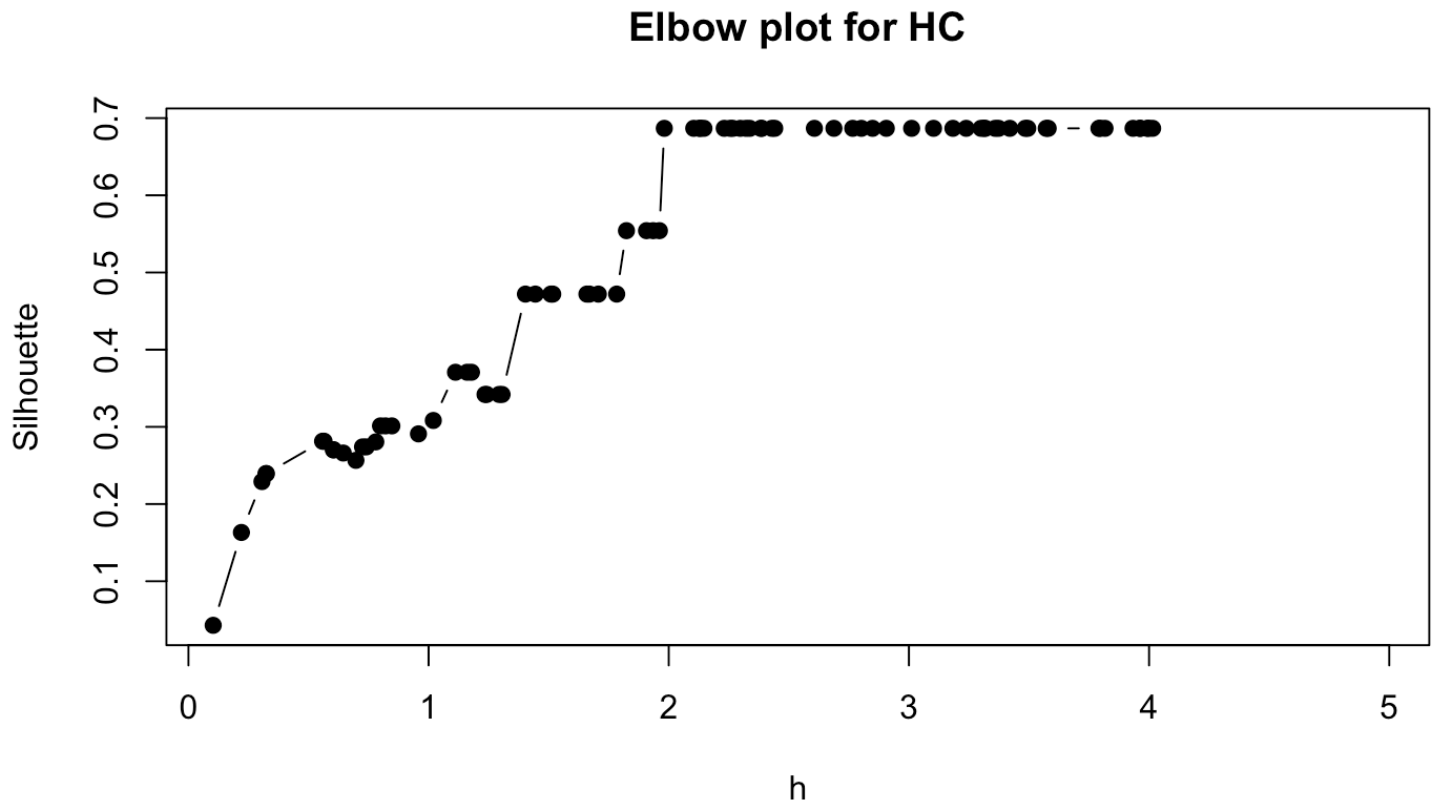
Elbow plots for silhouette and K

Do not need to recompute! Same dendrogram.



Elbow plots for silhouette and height

Do not need to recompute! Same dendrogram.



Other clustering methods not covered

Model-based clustering:

- **Mixture models**: Fit mixtures of distributions to data using the EM algorithm; provides soft cluster assignments (probabilities) and statistically principled model selection (BIC).
- R: `Mclust` (package `mclust`).

Other clustering methods not covered

Density-based clustering:

- **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise): Finds clusters of arbitrary shape based on local density; automatically detects noise/outliers; does not require specifying K in advance.
- R: `dbscan()` (package `dbscan`).

Other clustering methods not covered

Graph-based clustering:

- **Spectral clustering**: Uses eigenvectors of the graph Laplacian to embed data in low-dimensional space, then applies k-means; effective for non-convex clusters.

Other clustering methods not covered

Soft clustering:

- **Fuzzy c-means**: Assigns each observation a membership degree (between 0 and 1) to each cluster instead of hard assignment; useful when boundaries are uncertain.
- R: `cmeans()`

Others: Self-organizing maps (Kohonen maps), dynamic time warping for time-series clustering.

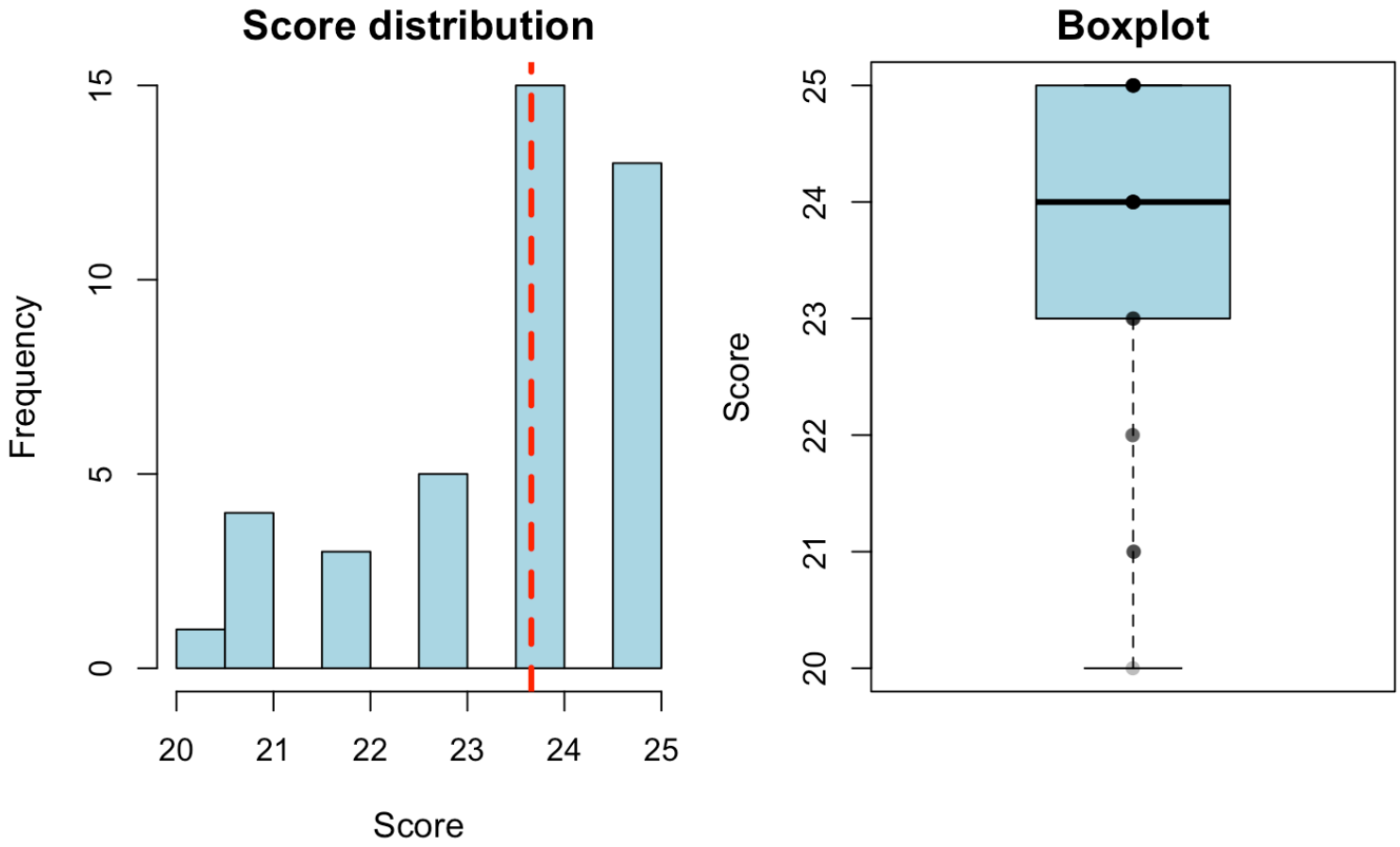
Assignment results

Overview

Assignment 1 – STAT340 (2026)

- Answers: 41
- Questions: 25
- Mean score: 23.66 / 25
- Correct rate: 94.6 %

Score distribution



How to run clustering here?

Which distance?

Which method?

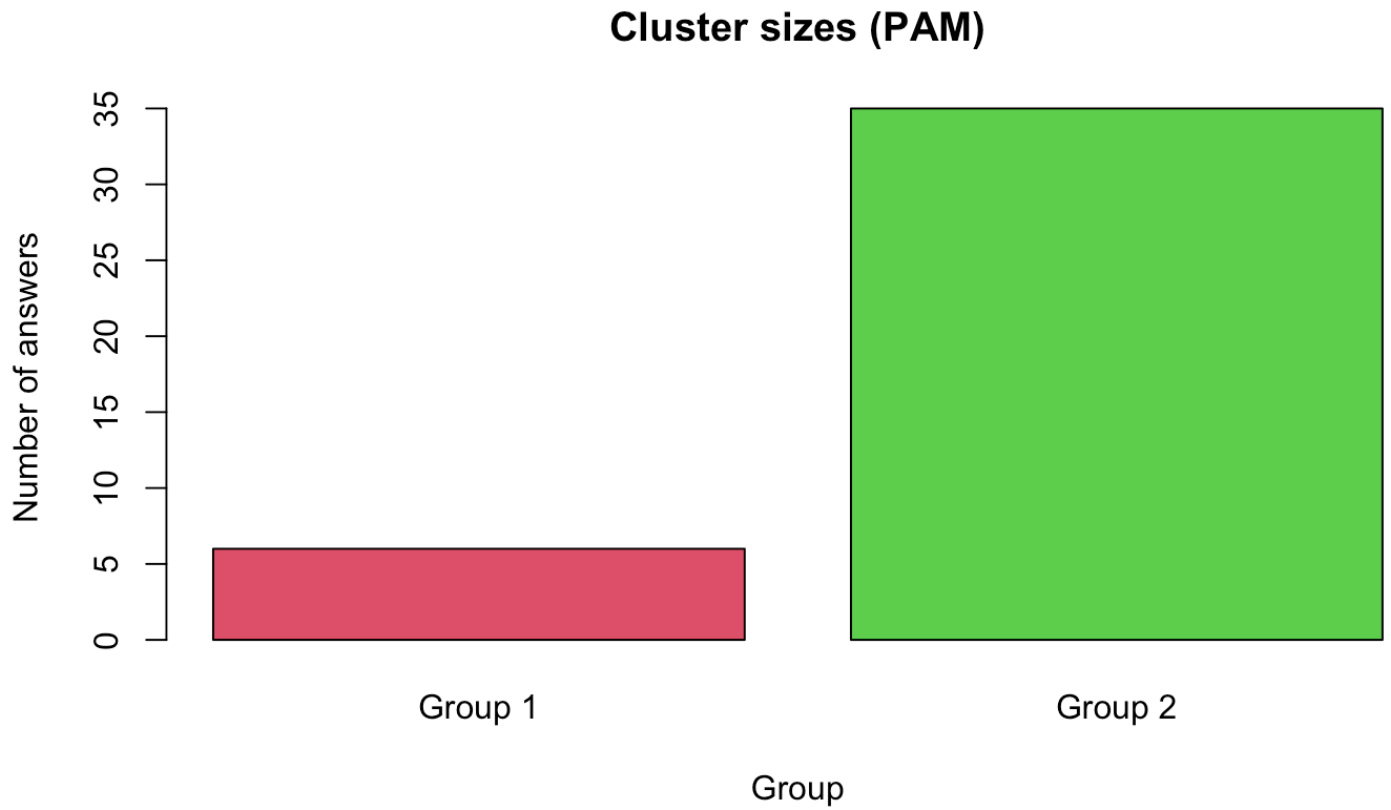
How to select the number of clusters?

Clustering setup

- Data: 25-dimensional binary vectors (correct/incorrect)
- Distance: binary Hamming distance (number of different answers)
- Method: PAM (Partitioning Around Medoids) on the distance matrix
- Number of clusters: chosen by an F-type criterion
  - For each k = 2, …, 8:
    - run PAM,
    - compute F-ratio = between-cluster variance / within-cluster variance (of the scores),
    - require at least 3 answers per cluster.
  - Choose k with the largest F-ratio.

PAM clustering

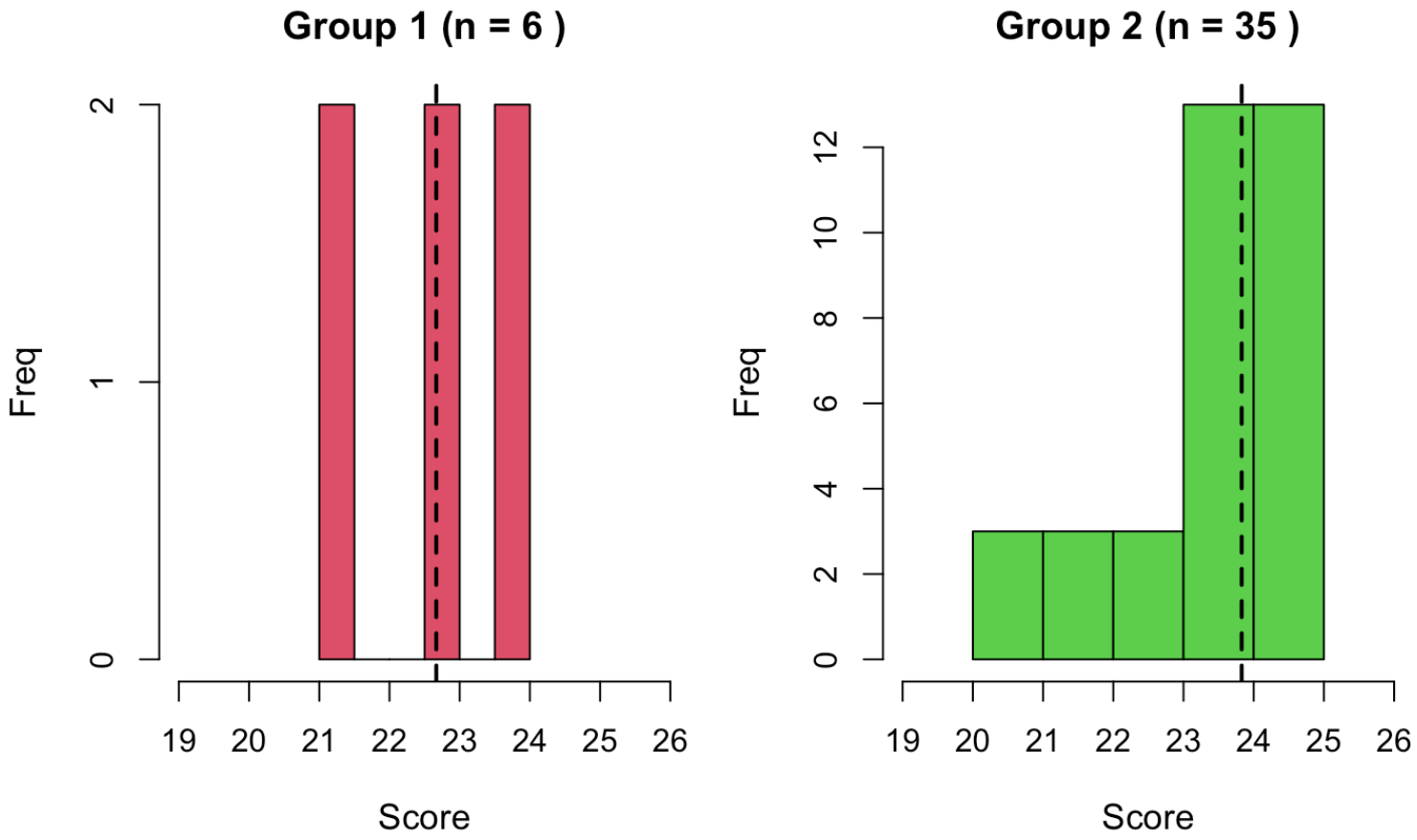
Optimal number of clusters (PAM): 2



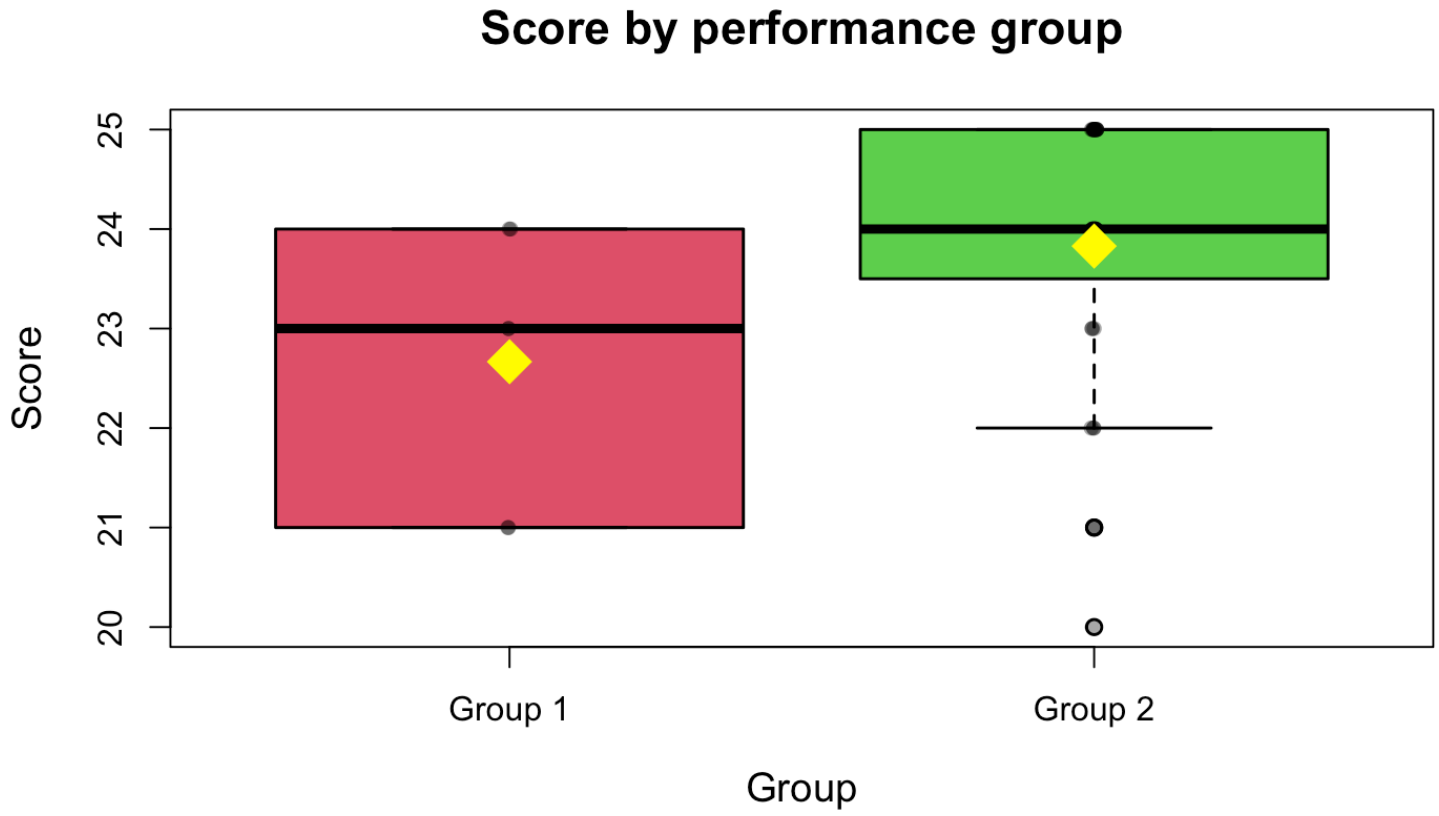
Cluster statistics

Group	N	Mean	SD	Min	Max
Group 1	6	22.67	1.37	21	24
Group 2	35	23.83	1.34	20	25

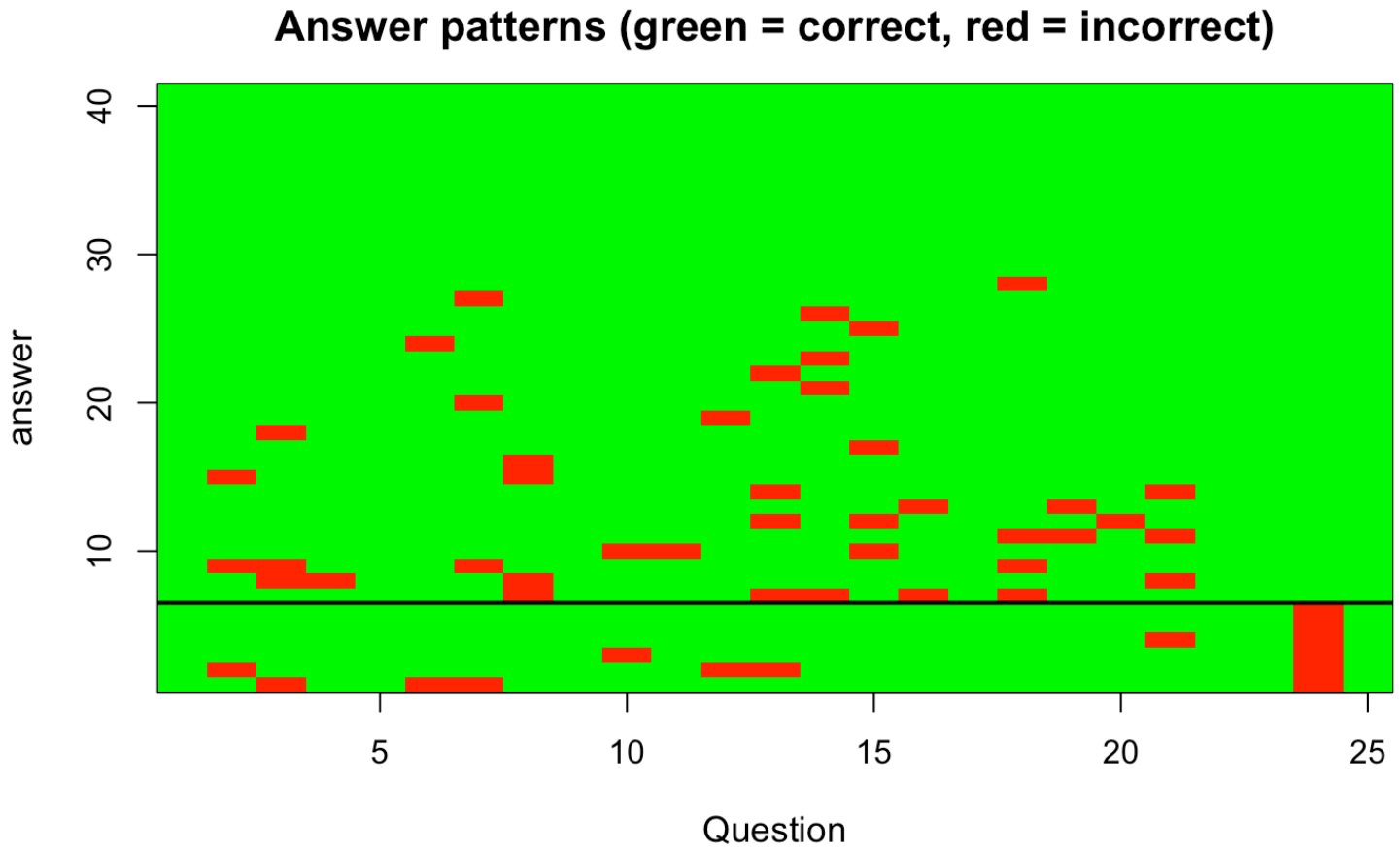
Score distributions by group



Boxplot by group



Answer pattern heatmap



ANOVA results

One-way ANOVA: score ~ performance group

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## performance  1   6.91   6.915   3.836 0.0573 .
## Residuals   39  70.30   1.803
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Result: F = 3.84, p = 0.057

Post-hoc: Tukey HSD

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = score ~ performance, data = results_df)
##
## $performance
##                     diff         lwr      upr     p adj
## Group 2-Group 1 1.161905 -0.03806934 2.361879 0.0573483
```

Summary

- Binary Hamming distance on 25-question 0/1 answer patterns.
- PAM clustering on the Hamming distance matrix.
- Number of clusters chosen by an F-type criterion with minimum cluster size 3.
- ANOVA and Tukey HSD show how groups differ in total score.
- What have we done wrong?

Assumptions